



Real Model-Based Systems Engineering Lab Manual

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[A companion to *Real MBSE: Model-Based Systems Engineering using LML and Innoslate®*](#)

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Real MBSE Course Syllabus

Week	Lecture Topics	Lab Manual	Individual Assignment	Team Assignment
1	Methodology and LML	N/A	Real MBSE Chapter 1, p. 1-14 LML Specification Section 1&2 Sign-up for Innoslate	N/A
2	Innoslate and Middle-Out Process	Getting Started	Real MBSE Chapter 1, p. 14-22 LML Specification Section 3.1-3.3 Innoslate website	N/A
3	Role of SE in Lifecycle	Forming Teams	Real MBSE Chapter 2	Form Teams
4	Program Management Processes	Sample Project	Real MBSE Chapter 3, p.35-44	Form Teams
5	Creating Project Work Plan	Lab 1	Real MBSE Chapter 3, p.45-51	Build Draft PMP
6	Requirements Analysis and Management	Lab 2	Real MBSE Chapter 4	Derive and Analyze Requirements
7	Functional Analysis	Lab 3	Real MBSE Chapter 5, p. 71, 72	Develop scenario model
8	Simulation	Lab 4	Real MBSE Chapter 5, p. 73, 74	Execute scenario model
9	CONOPS Development	Lab 5	Real MBSE Chapter 5, p. 76-94	Develop CONOPS scenario matrix
10	Solution Synthesis	Lab 6	Real MBSE Chapter 6	Identify potential implementations
11	Trade Studies	Lab 7	Real MBSE Chapter 7	Identify trade studies needed
12	Design and Integration Monitoring	Lab 8	Real MBSE Chapter 8	Design monitoring
13	V&V	Lab 9	Real MBSE Chapter 9	Create Test Suites
14	O&S/Disposal	Lab 10	Real MBSE Chapter 10 & 11	Identify O&S and disposal processes
15	Model-Based Review	Lab 11	Real MBSE Chapter 3, p. 44	Conduct an MBR of other teams work

Getting Started

LML has been implemented in a commercial tool called Innoslate®. Innoslate is a product developed by SPEC Innovations to integrate all the systems engineering and program management disciplines into a single tool. To get started with Innoslate, sign up for free at <https://cloud.innoslate.com/signup> with your .edu email address. You will be automatically upgraded to the Academic version, which gives you more entities to work with. It has all the same features as the complete cloud version. You may be placed into another Innoslate Organization for your class other than the general Academic Organization. Your professor will let you know if you need to change organizations and how to do this.

Once you have signed up for Innoslate, we recommend you explore the tool to get familiar with it, as you will be using it all throughout this course. At the top of the organization dashboard you will see a “Welcome to Innoslate” message. Select the “Start Guided Tour” for access to sample projects on how to use Innoslate (see Figure 1).

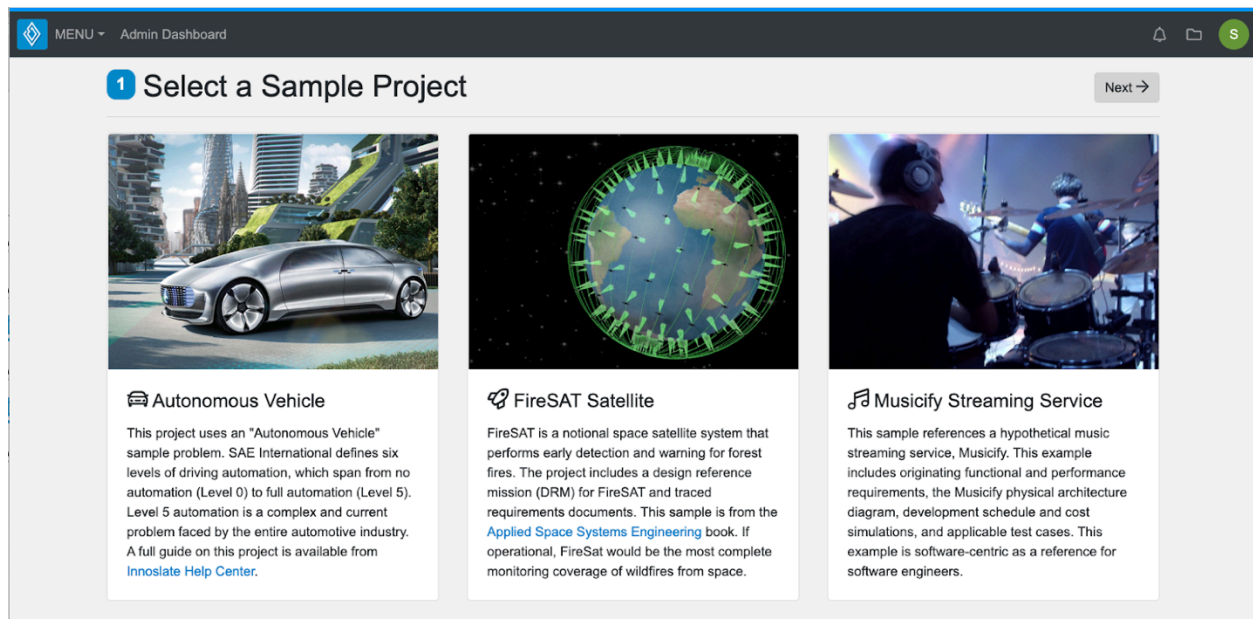


Figure 1. Sample Project Selection Screen

Select the Sample Project and then click “Next” and a screen like Figure 2 will appear.

You can customize your tour to focus on areas of interest. In preparation for one of your first lab projects, you might choose “Requirements Management” and “Data Management.” Note that this will automatically create a new project that you own with the sample. Selecting the “Next” button again starts the sample project tour. Keep selecting the “Next” button as shown in Figure 3.

Tip: If you want to follow along with the tour, you can create another new (blank) project and then view it in a separate browser tab. Remember that you are using a web tool, which means you can have multiple browser windows into the same Innoslate project. You will find this an interesting capability when you begin to build complex models of your system.

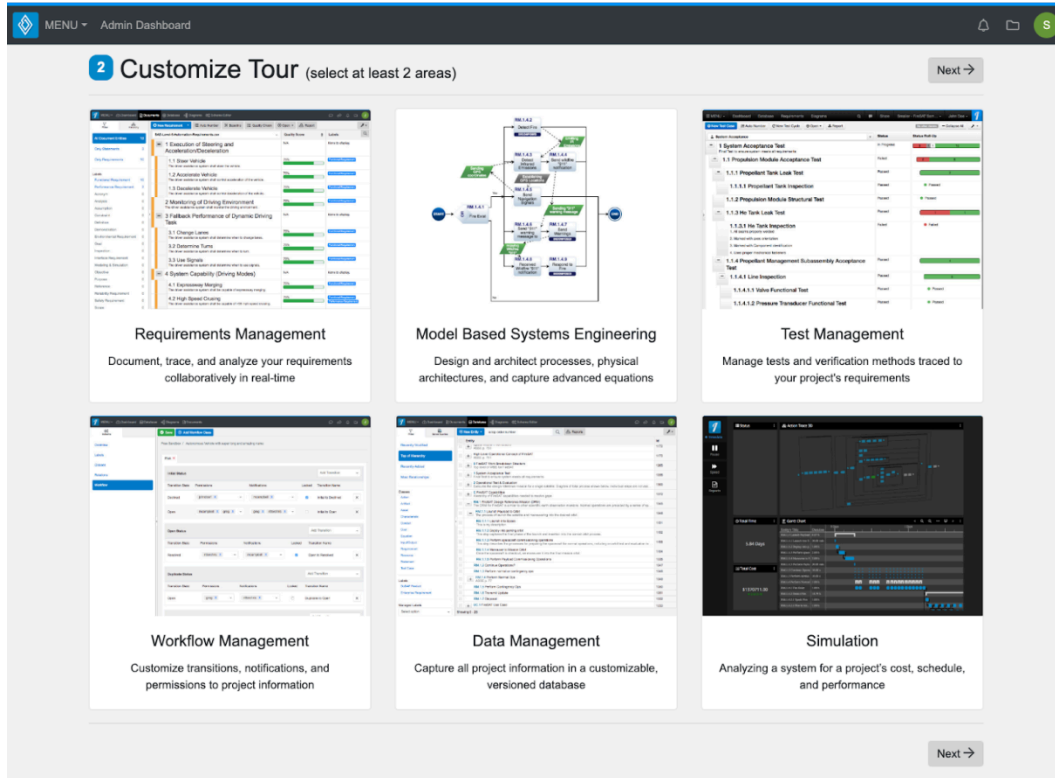


Figure 2. Selecting the tour domains.

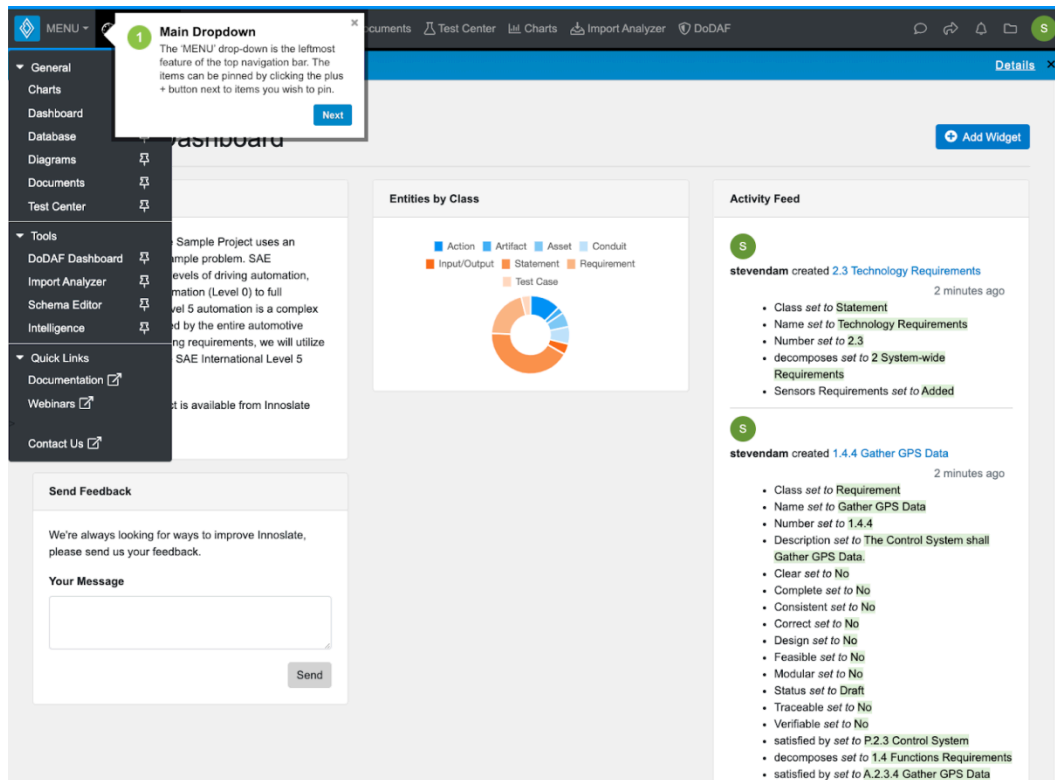


Figure 3. Beginning the tour.

Innoslate may be a bit overwhelming at first. Use the capability to pin and unpin information on the menu bar (as shown in Figure 3). You can also customize the dashboard for your needs. Add or remove the dashboard “widgets” as desired.

The Real MBSE book goes through each of the primary views of the tool. We recommend also following those pages in the (p.15-18) as well, to get familiar with the tool. The Innoslate website’s resources page provides links to videos and the Help Center to help you get started.

Tip: User support is free. If you need help getting started or have a technical question/concern, please contact support@innoslate.com. Also, if you have a good idea for a tool improvement, please use the Feedback form on the main dashboard. We welcome your input. If you like, we can provide your class with a demonstration. Please have your professor contact sales@innoslate.com for a free demonstration to the entire class. DO NOT get frustrated with the tool. We will help you as much as we can in getting started and using the tool, however we WILL NOT do your homework ;) That would defeat the purpose of your learning how to use the tool yourself.

Forming Teams

As a systems engineer you will often be part of a team, if not leading some or all that team. Since a systems engineer's job is to optimize the design engineering aspects of the lifecycle, you will be working with people from many disciplines. You will have to learn their "language" which includes the acronyms they use as shortcuts. If you are an undergraduate student in systems engineering, you may never have had any experience with other disciplines. It will help you significantly in your career if you at least get to know other engineers within the school. Try to use some of your electives to take classes in electrical or mechanical engineering or any other discipline you can. Likely you will have to take a physics and math class every semester taught in other Departments. Use these classes as an opportunity to make friends in the other disciplines and learn their language.

If you are a graduate student taking this course, you may find that you have had a variety of experiences with people from other disciplines, so you know why understanding other languages is very important. Use that experience in this class as well.

Of course, since you will be forming a team within the class to conduct the labs, you will be limited to the breadth of experience of your classmates as well. Spend time at the beginning of the semester getting to know each other. Find out who you can work with and who you can't (yes there will be people you do not want to spend the long hours associated with a course like this). When forming a team, compatibility is desirable, but not required. In reality, when you are on a job, you will be similarly limited by your officemates as well. And you may not get any choice in who you work with, short of finding another job (where the grass will likely not be greener, really).

Consider forming a team as your first trade study. As you will learn later, the important part of starting a trade study is identifying the criteria or metrics you will use to make decisions. In this case, we want to decide who we need on the team. Several factors can be used, and your professor may have a number of specific ones in mind. Follow those. They may choose the team for you, then it's "easy." You just have to figure out how best to deal with what you have.

If the professor leaves the teaming criteria up to you, then these criteria might be helpful.

Potential Teaming Criteria

1. **Schedule.** Identify each potential team member's schedules to see if there are times when you can work directly together. You also want to make sure there are times you can work separately. You need both. Innoslate will help with the collaboration, as you can work on the same project at the same time. If you work at different times, then the Activity Feed on the main dashboard will tell you who did what, when. Use this and the other collaboration features to work together at any time.
2. **Skills.** This criterion will have several sub-criteria. For this class, we don't really need experts in the design engineering disciplines, but it helps to understand something about the limits of the design engineering capabilities so you develop a feasible system design. Identify who understands other disciplines and how this knowledge can be beneficial. Within Systems Engineering, you will need people who can: analyze the current state of the system; derive and analyze requirements; model the system processes; simulate the processes; and describe how you can validate that the system meets the requirements. These require skills that vary from people who can:
 - a. See the big picture
 - b. Understand the details
 - c. Draw understandable diagrams

- d. Write clearly
 - e. Speak clearly and effectively.
3. Personality. As mentioned above, being able to work with others is an important skill. You need to determine your “type.” Are you introverted or extroverted? Do you prefer to work alone or in a team? Are you a leader or a follower? There are major systems, such as Myers-Briggs, for determining your unique personality type. You do not have to go to that extreme, but you do want to put together a group that can work together effectively. Spend time getting to know each other’s strengths and weaknesses in this area as well.

You may determine you need other skills, but these are some of the most critical. Use these criteria to determine the strengths and weaknesses of each team member. Be honest with your assessments. Encourage constructive criticism.

This crash course in soft skills may seem overwhelming at times, but an essential skill for the systems engineer is to know how to work effectively in a team. As you grow in your career you will likely be tasked to put together and lead a team. If you do not know much about these soft skills, don’t worry you will learn throughout your systems engineering career.

Sample Project

MBSE can be used in any number of industries. A fictitious project was created to put focus on the systems engineering itself rather than the details of the project. You can download the start of the project [here](#) and follow along.

The sample project that we used for this book and the labs is one of great current interest today (circa 2020): establishing a permanent base of the Moon. Ever since Apollo 18, 19, and 20 were cancelled, which were to establish a permanent base on the Moon, the United States and other countries have desired to go back to the Moon and use its resources to benefit the world. With the recent development of lower cost launch capabilities by billionaires, such as Elon Musk and Jeff Bezos, such a dream appears nearer reality. But how do we make sure that this new “colony” will survive and grow? We use systems engineering to take into account all the factors needed for a successful, long-term project. This work will require understanding the full lifecycle of the system, which would include the transportation, communications, energy sources, mining capabilities, environmental factors, safety, security, and many other factors. The book and the sample project provided just gets that project started. You, the student, will expand upon what has been provided and take it further.

We recommend that this class be used to further develop what would really be needed to establish such a base. If you can get a copy of Ben Bova’s book: “Welcome to Moonbase,” we highly recommend it as he thought through many of the aspects of such a development and operations, as well as presenting it in a very entertaining form. Obviously, that book was written in 1987, so the actual events did not come true anywhere close to what he projected, but the types of things that need to happen will be similar, if not the same.

Professors, we also recommend that as classes explore the different aspects of the architecture and design, you may want to pass your results on to the next class and have them build upon the previous class. This would be a way to create a series of papers for publication in various journals and help inform these government organizations and entrepreneurs how valuable systems engineering would be to all their endeavors. You may want to create your own project to be developed in parallel with this lab manual. The basic steps should be the same.

Lab 1. Build Draft Project Management Plan (PMP)

Tip: Begin by creating a work plan using a template. In the case of this sample problem, it is appropriate to use the NASA project management plan template provided in Innoslate.

Create a PMP Document

1. Select the Documents tab from the top toolbar.
2. Click on  .
3. In the dialog box, add the following properties
 - a) Type: “Project Management Plan Document”
 - a) Number: “PMP.1”
 - b) Name: “PMP for Moonbase”
 - c) Description: “Architecture-level project work plan”
- b) Click on the blue “Next” button.
- c) In the template dropdown menu, select “NASA Project Plan (With Guidelines).”
 - o *Note: This template provides a standard NASA Project Plan with the guidance for completing the plan provided to the user.*
- d) Click the blue “Finish” button. You should have the following results:

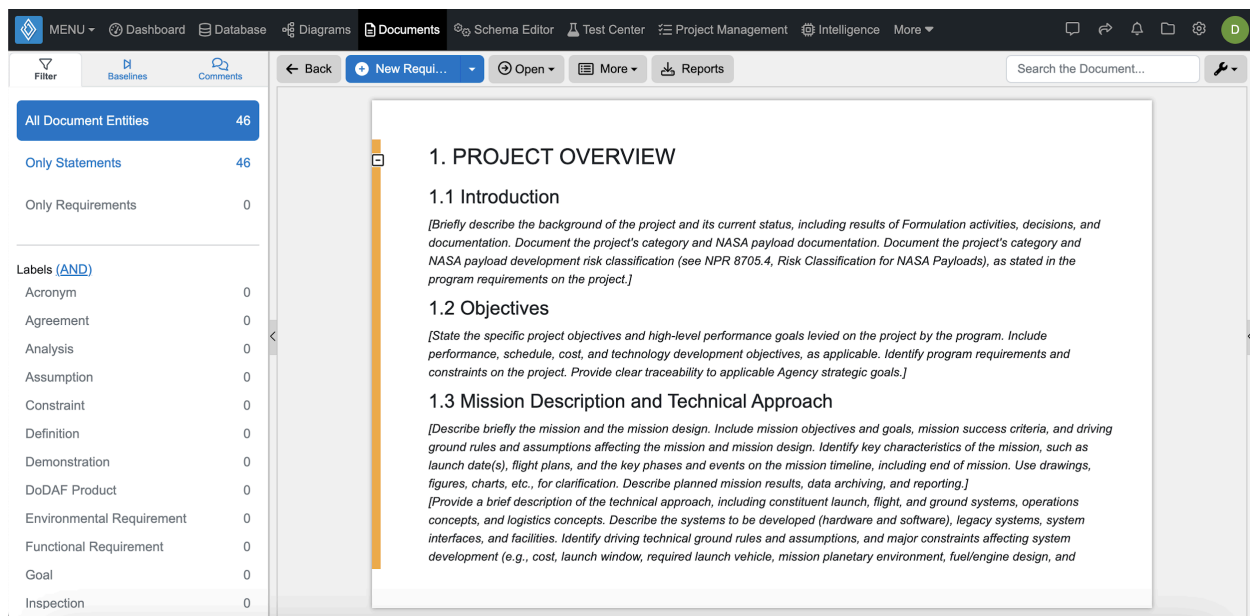


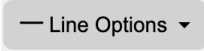
Figure 4. NASA Project Plan Template in Innoslate®

Create an Action Diagram

The next section covers step one of the processes shown in Figure 21 of [Real MBSE](#), “Develop Technical Scope Estimates.” The technical scope is often defined as the project’s context diagram. LML would suggest an Asset Diagram for this purpose.

1. Click the ‘Diagrams’ tab on the top menu bar
2. Select the  button.
3. Select “Asset Diagram” from the pull-down menu then click next.
4. Add the following attributes:
 - a) Number: “P.1”
 - b) Name: “Selene Project Context”
 - c) Description: “Provides a context diagram for the Moonbase: Selene project.”
5. On the left side panel, under the “New” tab, select and drag a new Asset entity onto the palette, placing it in the middle of the panel.
6. Give this new Asset entity the following attributes:
 - a) Number: “P.1”
 - b) Name: “Selene Project Context”
- e) Add the following new assets entities using the same steps as before:
 - a) Number – “P.1.2” Name – “Public”
 - b) Number – “P.1.3” Name – “Launch Services”
 - c) Number – “P.1.4” Name – “World Governments”
 - d) Number – “P.1.5” Name – “Workforce”
 - e) Number – “P.1.6” Name – “Funding Sources”
 - f) Number – “P.1.7” Name – “Construction Materials”
 - Note: You can always group Assets and create a decomposition of the group.

Link the Project to each stakeholder using a Conduit

1. Select the “Project Selene” Asset box.
2. Drag any of the green circles to the “Funding Sources” Asset.
3. Name the line “Status Reports.”
4. While still having the ‘Status Reports’ conduit selected, click the  button on the toolbar.
5. Click “Directed” under the Item Flow category in the dropdown menu.
6. Repeat steps 1 and 2 to create another conduit.
7. Name the new line, “Project Funding.”
8. While still selecting the ‘Project Funding’ Conduit, click the  button on the toolbar.
9. Click “Reverse” under the Item Flow category in the dropdown menu.

10. Once finished, click the green “Save” button in the upper left corner to save your work.

o *Note: Your diagram should look like the figure below.*

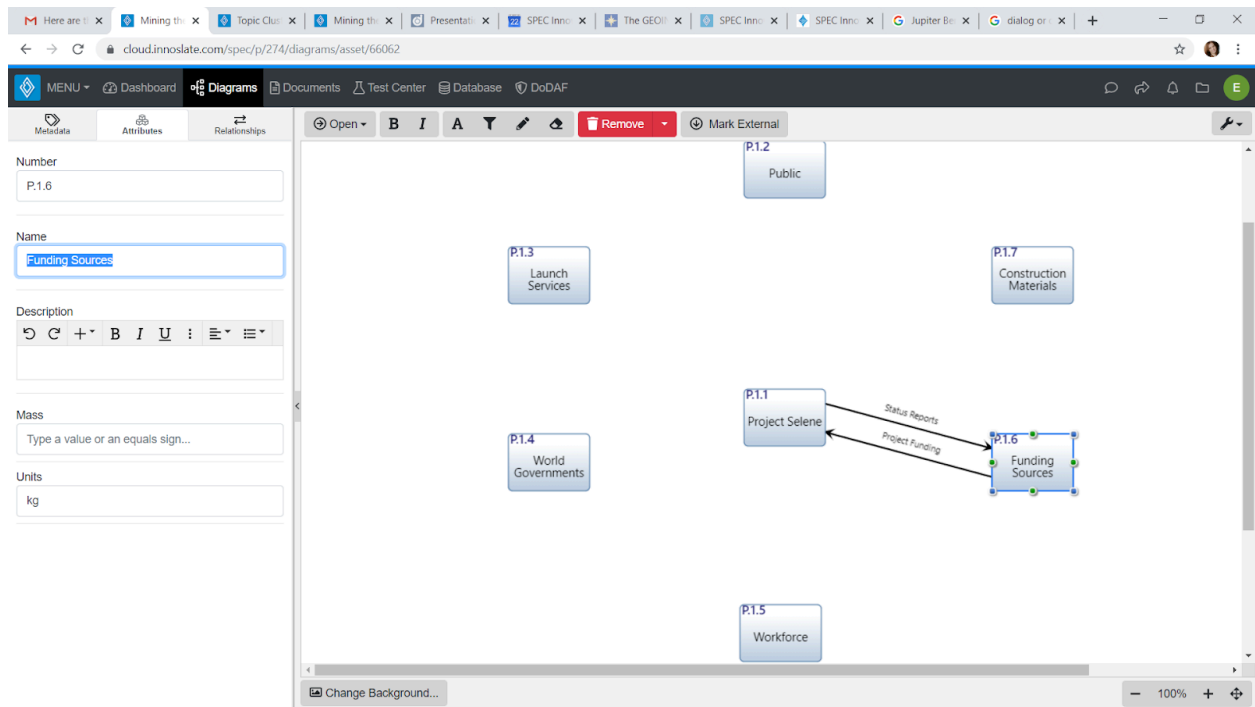


Figure 5. Starting the Context Diagram

Insert the Context Diagram into the PMP document

1. Navigate to Documents View from the top menu and select the document, “PMP for Moonbase.”
 2. Scroll down to Statement 1.5, “Stakeholder Definition.”
 3. Click on the statement to open the editing menu.
 4. In the editing menu, click the  dropdown button and select “Diagram.”
 5. Select the following options:
 - a) Diagram Type: “Asset Diagram”
 - b) Target Entity: “Selene Project Context”
- o *Note: Your results should be similar to the image in Figure 6.*

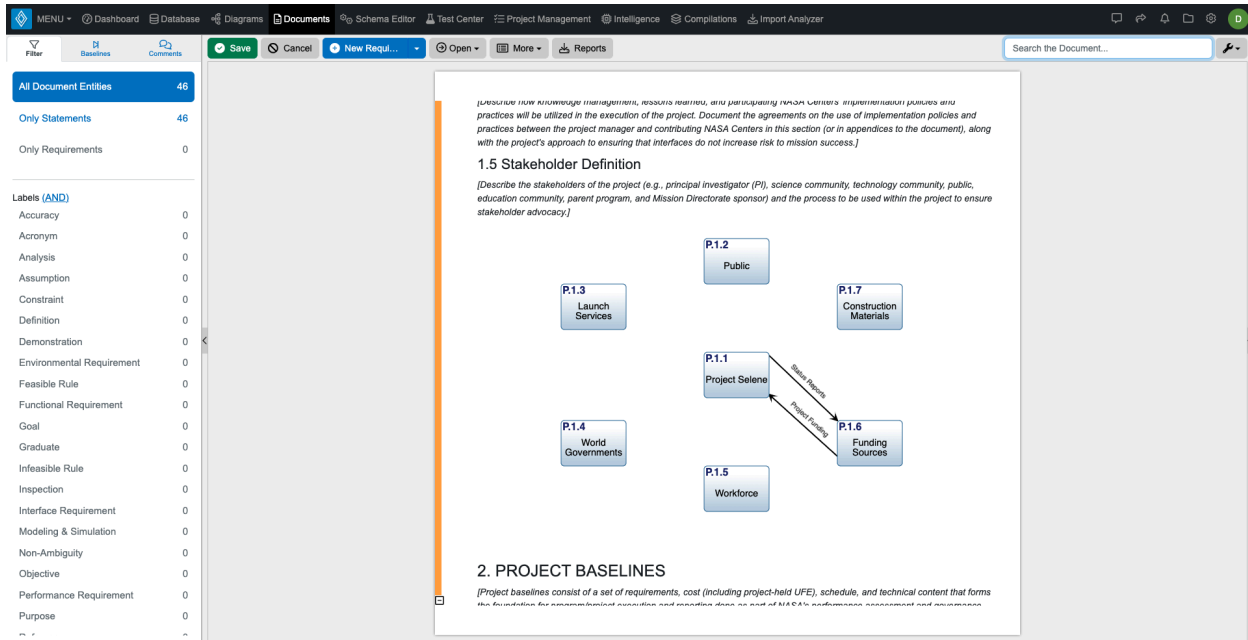


Figure 6. Adding the Context Diagram to the Program Plan

As you continue to refine the diagram, changes will be automatically updated in the document. You can even include the diagram's URL to allow the user to right-click on the diagram and open it up in a new tab. Any additional changes should be made in the diagram as the document will reflect those changes. This concordance is an essential attribute of MBSE.

Add Information to the Project Plan

The next step is choosing the process model and processes. The incremental lifecycle model (as shown on the right side of Figure 15 in the Real MBSE book) will be selected keeping in mind that each phase of the project may be revisited to update the plan due to decisions made as a result of performing technical activities.

1. Scroll down to Statement 3.6 in the PMP document.
2. Select the Statement and type into the description, "We will use an iterative, incremental lifecycle model. Each phase will represent a different increment, where this plan will be reviewed and adjusted as needed."

Create a Project Timeline Diagram

The next step is identifying the technical activities. Activities are captured as Actions in LML with an Action or Task type.

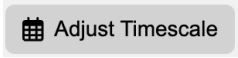
1. Click “Diagrams” from the Navigation Menu.
2. Click the  button.
3. Scroll down and select “Timeline Diagram”.
4. Add the following attributes:
 - a) Number: “SCHED.1.”
 - b) Name: “Moonbase: Selene Top-Level Development Schedule”
 - c) Description: “This schedule identifies the primary phases of development for the Moonbase.”
6. On the editing bar, click the  button.
7. Set the timescale to the following attributes:
 - a) Start Date: “1/1/2020”
 - b) End Date: “12/31/2030”
 - o *Note: The timescale may auto-adjust to a similar date.*
8. Click the “Class Type” dropdown under the “Timeline” and “Milestone” entity elements on the left menu bar. Choose “Task” as the entity type class.
 - o *Note: The default class for the Timeline Diagram is set to “Action.” We are changing the default from “Action” to “Task.” This will prove useful later in the lab.*
9. Drag a “Timeline” entity to the empty palette.
10. Add the following attributes to the entity:
 - a) Number: “SCHED.1.1”
 - b) Name: “Phase 0: Project Planning”
 - c) Description: “During this phase, we will develop the initial plans for the Moonbase: Selene project.”
 - d) Start: “1/1/2020 12:00:00 AM”
 - e) Duration: “=triang.dist(6, 12, 9) months”
 - o *Note: Child timelines can be created by dragging a Timeline entity over to its parent Timeline’s triangles. Continue adding timeline entities to your Timeline Diagram using the table provided below in Table 1. After all entities are added, the diagram should look similar to Figure 7 below (start times and durations may vary).*

Table 1: Table for Schedule

Number	Name	Description	Duration	Start	Milestone?	Task?
	Plan Approved		1 hour	10/10/2020 12:00 AM	X	
	Ready for Safe Operations		1 hour	2/20/2024 12:00 AM	X	
	Initial Delivery of Raw Materials		1 hour	11/8/2026 12:00 AM	X	
	Initial Delivery of Construction Materials		1 hour	1/18/2031 12:00 AM	X	
SCHED.1	Moonbase: Selene Top-Level Development Schedule	This schedule identifies the primary phases of development for the moonbase.	1 hour	N/A		X
SCHED.1.1	Phase 0: Project Planning	During this phase we will develop the initial plans for the Moonbase: Selene project.	=triang.dist(6, 12, 9) months	1/1/2020 12:00 AM		X
SCHED.1.2	Phase 1: Survey and Preparations	Conduct initial site survey mission.	3 years	1/22/2021 12:00 AM		X
SCHED.1.2.1	Initial Site Survey Mission	Survey to verify location for Moonbase: Selene.	12 months	3/26/2021 12:00 AM		X
SCHED.1.2.2	Site Preparation Mission		12 months	4/23/2022 12:00 AM		X
SCHED.1.2.3	Initial Operations Mission		1.307 years	10/5/2022 12:00 AM		X

Number	Name	Description	Duration	Start	Milestone?	Task?
SCHED.1.3	Phase 2: Initial Mining Operations	Mission to prepare the site for initial operations.	4 years	10/14/2022 12:00 AM		X
SCHED.1.3.1	Raw Material Extraction Facility Mission		1.775 years	10/19/2022 12:00 AM		X
SCHED.1.3.2	Raw Material Transportation System Mission		3.354 years	6/14/2023 12:00 AM		X
SCHED.1.3.3	Expanded Habitat Mission		3.773 years	1/26/2023 12:00 AM		X
SCHED.1.4	Phase 3: Final Mining Operations		5.45 years	8/11/2025 12:00 AM		X

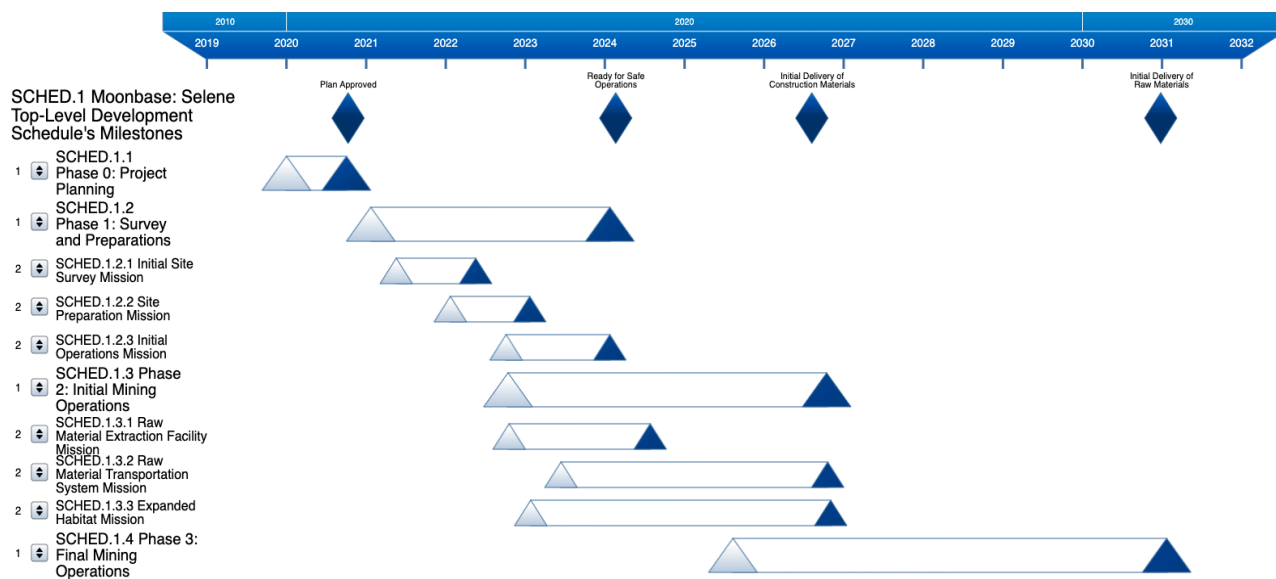
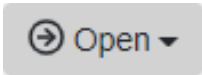


Figure 7. Completed Timeline Diagram

When creating Task or Action entities on the Timeline diagram, all the entities, attributes, relationships, and metadata will be available to work with. It is always recommended to fill in all known attributes to create the most complete description of the entity. In this case, the known attributes are the number, name, description, start date, and duration.

The Start Date of an entity can be set either in the attributes on the sidebar or by dragging the triangle on the diagram directly. You can also set the duration by dragging the blue circle on the side of the triangle to the right when the entity is selected. For projects like this one, distributions can be used to reflect the uncertainty in the duration of a task or action.

Create an Action diagram from the Timeline Diagram

1. While still in the timeline diagram, select  from the top menu bar and click “Action Diagram.” The resulting Action Diagram should look like Figure 8 below.

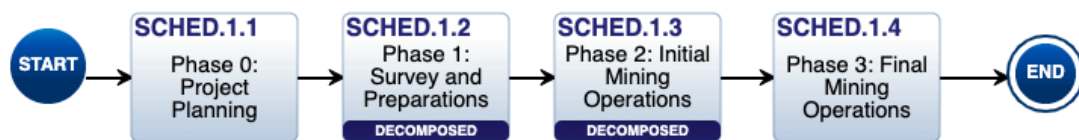
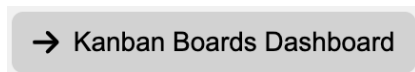


Figure 8: Action Diagram for Timeline

Create a Kanban Board for Project Management

Once all tasks have been declared in the timeline diagram, they can be put through the project workflow to evaluate the overall progress of each task, assign users, or organize tasks.

1. Navigate to the “Project Management” view from the top menu bar.
 - o Note: From the project management dashboard, users can view the project calendar, track upcoming due dates, and much more. These metrics are viewed through widgets. More widgets can be added via the “Add Widget” button.
2. Click the  button.
3. Click on  and select Project Workflow Board as the template.
4. Add the following attributes:
 - a) Number: “KB.1”
 - b) Name: “Moonbase: Selene Top-Level Kanban Board Tasking”
 - c) Description: “This Kanban board identifies the tasks and categorizes them by status for the Moonbase.”
5. Select the “1 To Do” column. Then click  and choose “Relate Existing Task.”
6. Include all tasks from SCHED.1.2 to SCHED.1.4.

7. Select the “2 In Progress” column. Using the “New” and “Relate Existing Task” buttons, add the task “SCHED.1.1” to be in progress.
8. Once all tasks have been added, click on “SCHED.1.1” and add the following attributes on the left sidebar:
 - a) Percent Complete: “15%”
 - b) Status: “In Progress”
 - c) Assignee: *Select your username*
 - o *Note: Your Kanban Board should look like Figure 9 below.*

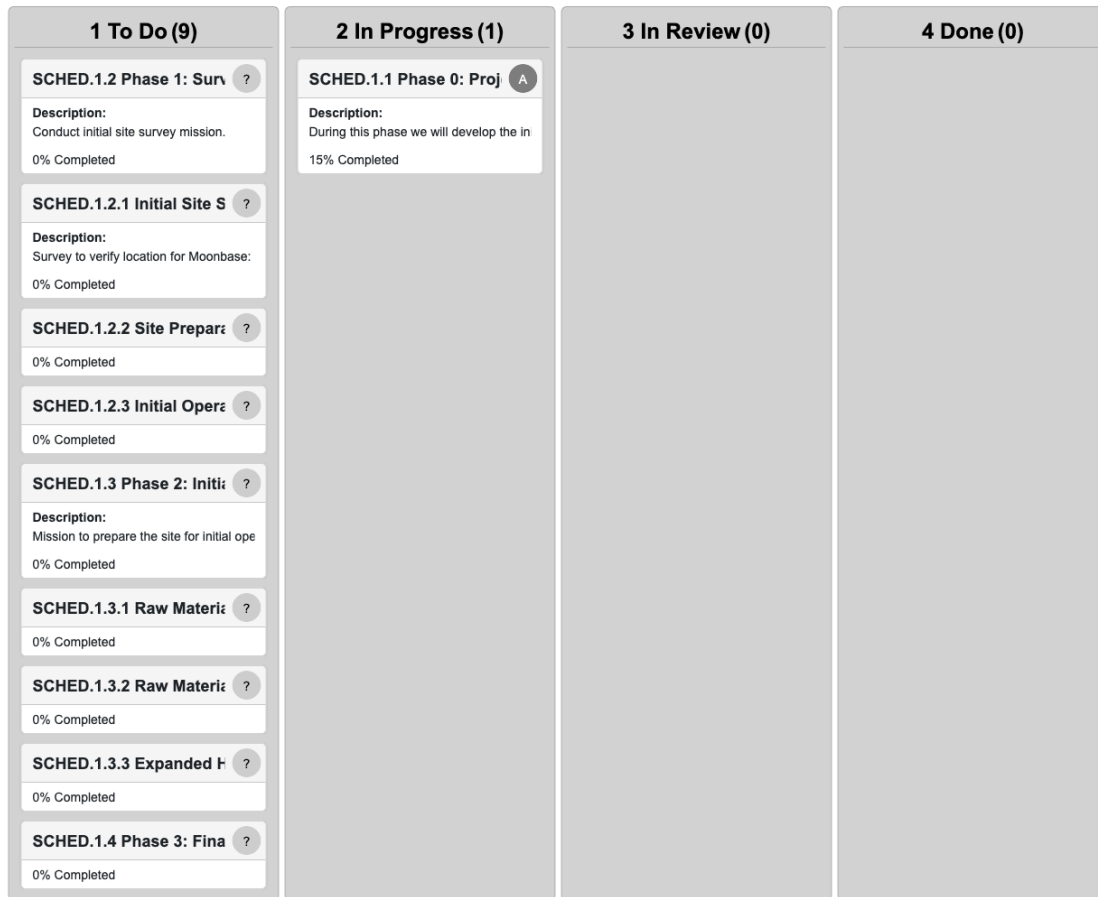
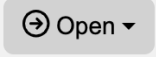


Figure 9: Kanban Board based on Tasks in the Timeline

This completed demonstration is an example of how the Kanban Board can be used to create tasking boards for engineers to use in Project Management. The tasks on the Kanban Board have the flexibility to assess Percent Complete, Due Date, Estimated Time of Completion, Finish Date, Status, and Assignee.

Create a Gantt Chart from the Kanban Board

Gantt charts are an essential part of project management. Innoslate can automatically generate a Gantt chart from an existing timeline diagram & Kanban board with tasks.

1. While still in the Kanban board, click on  then scroll down to select “Gantt Chart.”
2. Create a dependency by clicking on “SCHED 1.2.1” and selecting .
 - o Note: The entity will be highlighted in a yellow border.
3. Click on “SCHED 1.2.2” to create the dependency.
4. Ensure that entity “SCHED 1.2.1” has “Finish” selected and entity “SCHED1.2.2” has “Start,” then click “Add.”
 - o Note: This will create a relationship between the two entities of “depends on.”

Add diagrams in Documents View

Finally, add the timeline to the PMP document created.

1. Open the PMP Document (from Documents view) and scroll to section 2.3.
2. Select the section, click the insert () button, and select Diagram from the menu.
3. Select “Timeline Diagram” for the Diagram Type and then “Moonbase: Selene Top-Level Development Schedule” for the Target Entity. The result will look similar to Figure 10 below.

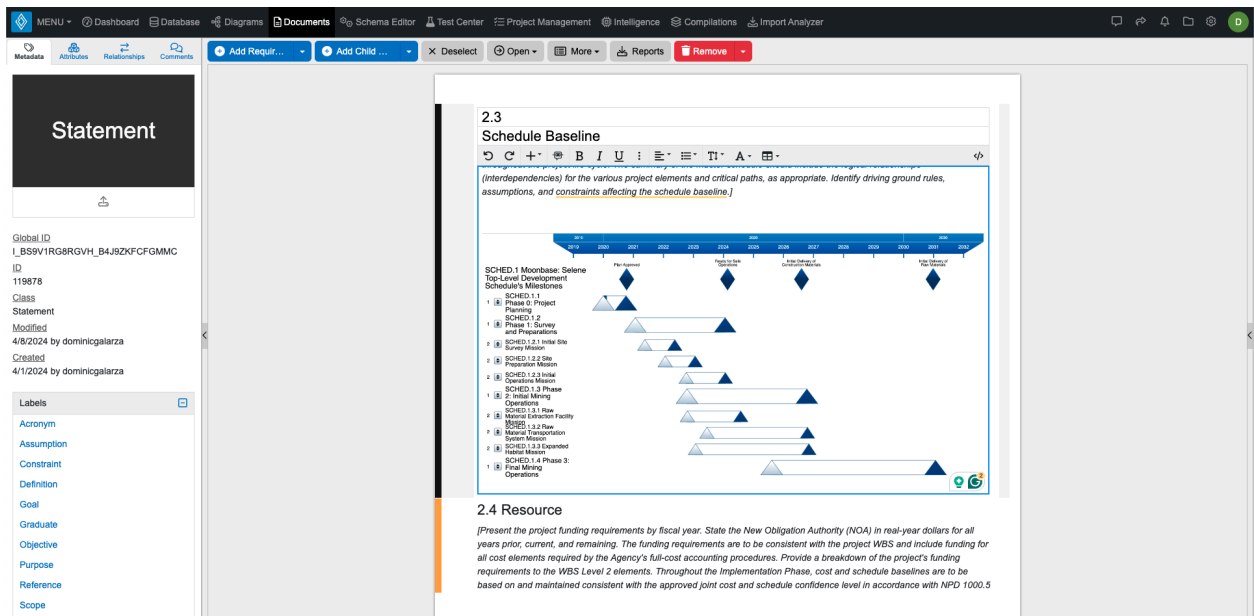


Figure 10: Timeline Diagram inserted into PMP Document

Lab 2. Derive and Analyze Requirements

[Welcome to Moonbase](#) was written as a manual for new lunar pioneers who had just arrived on base. There are two options to develop a set of requirements from the book: 1) read the entire book carefully and extract a set of requirements from it, or 2) use the book as a basis to create a set of scenarios or use cases to derive the functional requirements. The second approach is more applicable to the processes shown in the next chapter. Therefore, in this lab, the focus will be on the first approach.

Add an Attribute to a Class

The schema editor can add helpful properties to entity classes. For example, users can add referenced page numbers from Artifacts or other resources under the Statement class by applying the following steps. Tip: You must be the Owner of an Innoslate Project to use the schema editor.

1. Enter the Schema Editor and navigate to “Classes” on the left Menu Bar.
2. Scroll and select the “Statement” class.
3. Select .
4. Enter the following attribute information:
 - a) Attribute Name: “Page Number”
 - b) Type: “TEXT”
5. Push the green “Save” button at the top of the page.

Add a New Document Type to the Schema

For this project, user needs are captured as a “Statement” entity class. In this next part, a new document type must be created in the Schema Editor that allows a User Needs Document in Innoslate.

1. Enter the Schema Editor and navigate to “Labels” on the left-hand Menu Bar.
2. Click the  button.
3. In the “Create New Label” form, copy the information from Figure 11.

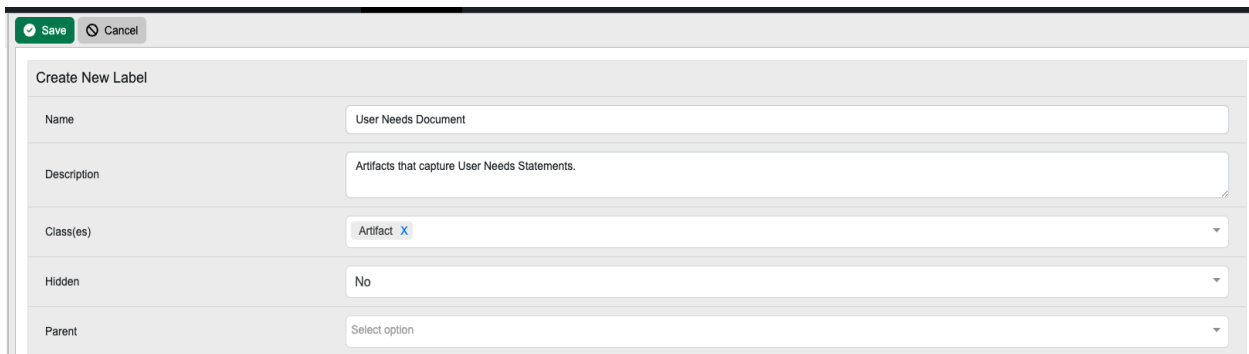


Figure 11: Identifiers in Schema Editor

Note: To make sure that the Document Type shows up in the pull-down menu, make sure that there is no "s" at the end of "Document" and that there are no spaces after the word "Document."

Apply the New Document Type

This process applies either Default or Customized labels to a document.

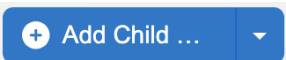
1. Go to "Documents View" from the top menu bar and click "Create a New Document."
2. Fill in the following attributes for the document:
 - a) Type: "User Needs Document"
 - b) Number: "UND.1"
 - c) Name: "Project User Needs Document"
 - d) Description: *Give a brief description*
3. Select "Next" then "Finish" to open the new document.
4. Click the "Back" button to exit the document.
 - a) *Note: For this project, the document "WtMB.1 Extracts from Welcome to Moonbase" has been provided. It is currently labeled as a "Requirements Document." The next few steps will be changing the document type to "User Needs Document."*
5. From the Documents Dashboard, locate "WtMB.1 Extracts from Welcome to Moonbase."



6. Highlight the document and click on the "Edit Entity" button () located at the top of the document.
7. In the "Labels" attribute, remove the current label of "Requirements Document" then click the dropdown menu and scroll down to select "User Needs Document."

Decompose and Transform Requirements in a Document

Top-level requirements or statements may need to be decomposed into additional entities. When a child entity is made, the "decomposes" and "decomposed by" relationships are automatically created and the numbering scheme will decompose along with the entity.

1. From the Documents View, open "UND.1 User Needs Document."
2. Click the arrow on the  button to open the drop-down and select "Requirement."
3. Click on  to add a child requirement.
 - a) *Note: Requirement entities can be transformed into statements and vice versa.*
4. Click on  to open a drop-down menu of options and select "Transform."
5. Select "Statement" as the Transform Class then click "Transform."

- o *Note: The document may take a few seconds and reload to save the change. Once this happens, the entire document is saved with any previous changes.*

Create Rationales using the AI Assistant

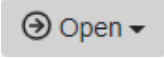
1. From the Documents View, open “WtMB.1 Extracts from Welcome to Moonbase.”
2. Scroll down and select the requirement entity, “WMB.1.1.6.2 Contain Oxygen.”
3. Click the AI assistant button () in the bar above the description field.
 - o *Note: The chatbot will appear on the right side of the screen. The AI Assistant is powered by Open AI’s Chat GPT, available to academic users within a limit of 1000 tokens per browser session. It is recommended that users prompt important and effective questions to their chatbot.*
4. The requirement has automatically been placed in the prompt section. Type, “What is a good one-sentence rationale for the mentioned requirement,” after the requirement sentence that the AI has already placed.
5. Evaluate the generated answer and copy the answer into the rationale section of the requirement.

Practicing with the Quality Checker

1. Open the “RT.1 Top-Level Requirements Document for Selene” document.
2. Click on  to open a drop-down menu of options and select “Quality Check.”
 - o *Note: The best score achieved by the quality checker alone is 75%. This is because two fields have to be manually checked:*
 1. *Correct*
 2. *Feasible**These two fields require human judgment, as the AI checker cannot determine the legality or feasibility of a requirement from text alone. Users can adjust the score on the left menu bar (under the Attributes tab) as correctness and feasibility are verified.*
3. Adjust the attributes of each requirement so that they achieve a 100% score, then save and exit the document.
 - o *Note: As the system develops further, you can trace these requirements to a Test Case or Action Diagram to show where they’re applied in the system.*

Create a Tree Diagram from a Document

Tree Diagrams can be used to visualize the hierarchy of chapters like Figure 12 below.

1. From the Documents View, open “WtMB.1 Extracts from Welcome to Moonbase.”
2. Select the Open button ().
3. Scroll down and click “Tree Diagram” on the Dropdown Menu.

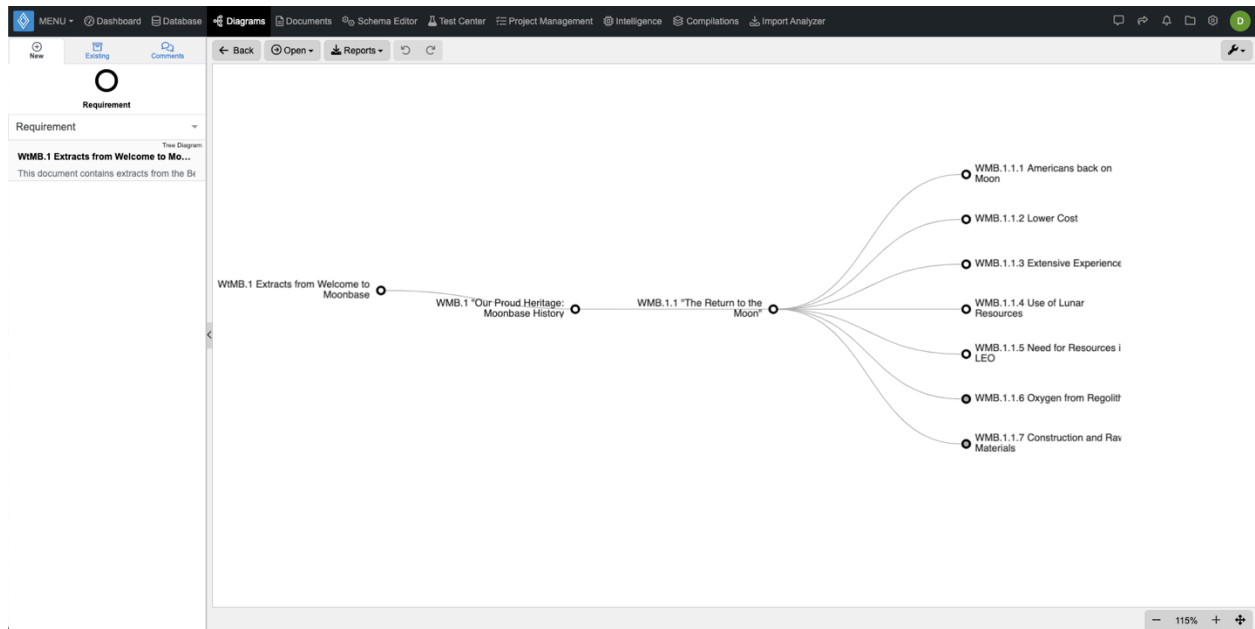


Figure 12: Tree Diagram

Notes on Tree Diagram

The first node in this diagram is the Artifact, “WtMB.1 Extracts from Welcome to Moonbase.” It shows a trace to the first Statement in the document, “WMB.1 Our Proud Heritage: Moonbase History.” The relationship between the two entities is “source of.” Next is the trace from “WBM.1 Our Proud Heritage: Moonbase History” to the child statement, “WMB.1.1 The Return to the Moon,” which is made through the “decomposed by” relationship, as well as the children of 1.1. These relationships are automatically created and managed by the tool.

Notes on Spider Diagram

The Spider Diagram shown below displays direct relationships between entities. To open and view the diagram, click on the WMB.1, “Our Proud Heritage: Moonbase History,” in the Tree Diagram. Use the “Open” button on the Menu Bar and scroll to the Spider Diagram to open the diagram.

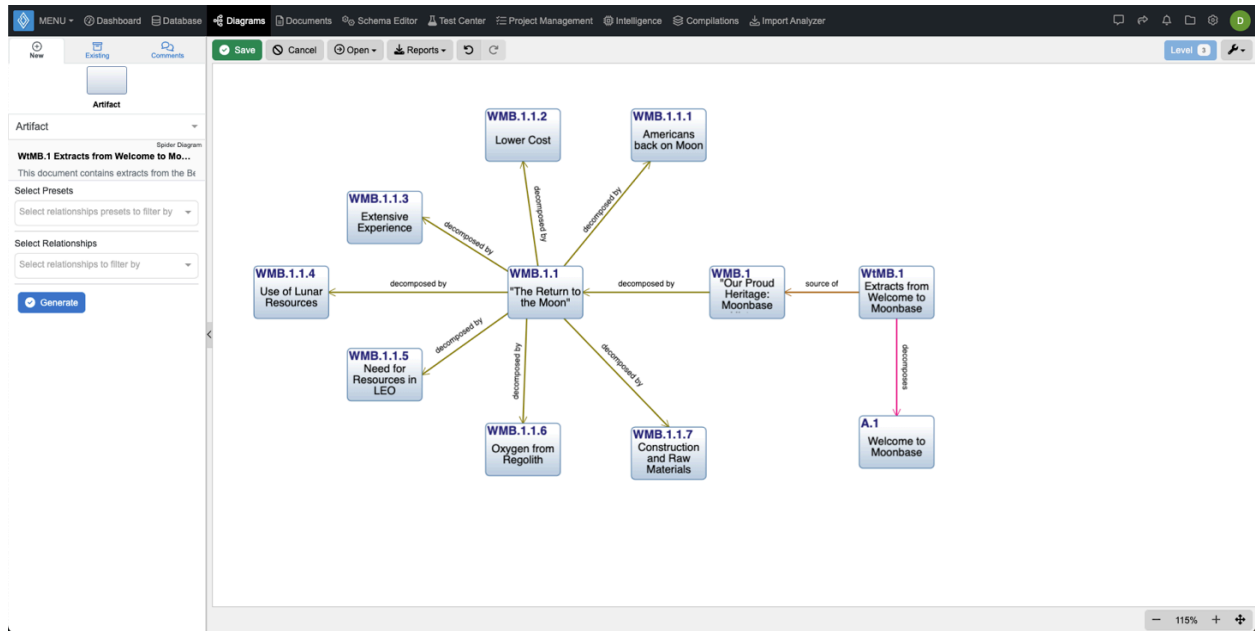


Figure 13: Spider Diagram

Add Decision and Issue Entities to the Project

As the project is developed further, the documentation and tracking of key project management decisions made by the customer and the team will be important.

1. Push the “Database” button on the Top Menu bar.
2. Press  and create a new “Decision” entity.
3. Fill in the following Attributes:
 - a) Number: “D”
 - b) Name: “Key Project Management Decisions”
 - c) Description: “Decisions on how we adjusted the schema and modeling to meet the specific requirements of this project.”
 - d) Justification: “Documentation of these decisions will aid in future development of the model through the lifecycle.”
 - e) Assumptions: “Idea is backed by all team members.”
 - Note: Your screen should look like Figure 14.

The screenshot displays the 'Decision' class entity view. On the left, a sidebar shows the entity's metadata, including its Global ID, ID, and class. The main area is divided into two panels: 'Attributes' and 'Relationships'.

Attributes Panel:

- Number:** D
- Name:** Key Project Management Decisions
- Description:** Decisions on how we adjusted the schema and modeling to meet the specific requirements of this project.
- Justification:** Documentation of these decisions will aid in future development of the model through the lifecycle.
- Assumptions:** Idea is backed by all team members.

Relationships Panel:

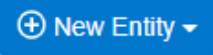
- alternative Statement
- causes Issue, Risk
- date resolved by Time
- decision due Time
- decomposed by Children
- decomposes Parents
- enabled by Many
- enables Decision
- equation of Equation
- incurs Cost
- located at Location

Figure 14. Decision Class Entity View

1. Press the green “Save” button to save and exit the Entity View.

Create a New Decision Entity in the Database

One of the first decisions that needs to be made is about the feasibility of creating construction materials on the Moon. Create a new Decision entity in the database that will decompose the parent decision entity, “Key Project Management Decisions.”

1. Push the “Database” button on the Top Menu bar.
2. Press  and create a new “Decision” entity.
3. Fill in the following Attributes:
 - a) Number: “D.1”
 - b) Name: “Feasibility of creating construction materials on the Moon”
 - c) Description: “We need to determine the feasibility of creating the kinds of construction materials needed for space stations in orbit around the Earth, Moon, and other locations.”
 - d) Justification: “This long-term requirement will drive costs and reduce the potential for return on investment, thus it must be resolved early in the lifecycle.”
 - e) Assumptions: “1. We know what kinds of construction materials will be needed by orbital platforms 10-20 years from the beginning of this program. 2. These construction

materials include shielding materials to reduce radiation exposure on these space-based platforms.”

4. Under the “Relationships” tab, locate the relationship, “*decomposes Parents*.”
5. Click the “Add” button then type and add, “D. Key Project Management Decisions,” to create the relationship.

○ *Note: Your screen should look like Figure 15.*

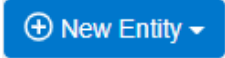
The screenshot shows the INNO SLATE database interface. The top menu bar includes options like Dashboard, Database, Diagrams, Documents, Schema Editor, Test Center, Project Management, Intelligence, Completions, and Import Analyzer. The left sidebar shows the 'Decision' entity with its metadata: Global ID, ID, Class, Decision, and Created date. The main area is divided into two tabs: 'Attributes' and 'Relationships'. The 'Attributes' tab is active, showing a form for the 'Decision' entity. The form fields are: Number (D.1), Name (Feasibility of creating construction materials on the Moon), Description (We need to determine the feasibility of creating the kinds of construction materials needed for space stations in orbit around the Earth, Moon, and other locations.), Justification (This long-term requirement will drive costs and reduce the potential for return on investment, thus it must be resolved early in the lifecycle.), and Assumptions (1. We know what kinds of construction materials will be needed by orbital platforms 10-20 years from the beginning of this program. 2. These construction materials include shielding materials to reduce radiation exposure on these space-based platforms.). The 'Relationships' tab is also visible, showing a list of relationships with 'Add' buttons.

Figure 15. Filled In Decision Entity

6. Press the green “Save” button to save and exit the Entity View.

Create a New Issue Entity in the Database

An “Issue” class entity can be created for the same purposes as the “Decision” class entity. However, “Status,” “Assignee,” and “Priority” attributes replace “Justification” and “Assumption.” It is recommended to use both classes, especially for major projects.

1. Push the “Database” button on the Top Menu bar.
2. Press  and create a new “Issue” entity.
3. Fill in the following Attributes:
 - a) Number: “1.1”
 - b) Name: “Feasibility of creating construction materials on the Moon”
 - c) Description: “Knowing whether or not we can construct valuable and useful materials that can support construction in space is critical to the success of this project.”

- d) Status: “Open”
 - e) Assignee *Insert your own username here*
 - f) Priority: “Critical”
4. Under the “Relationships” tab, locate the relationship, “*caused by Many*.”
 5. Click the “Add” button, then type and add, “WMB.1.1.7.2 Provide construction materials,” to create the relationship.
 6. Locate the relationship, “*mitigated by Many*.”
 7. Click the “Add” button, then type and add, “TS.1 Construction Materials from Moon Trade Study,” to create the relationship.

○ *Note: Your screen should look like this:*

Figure 16. Issue Entity View

8. Press the green “Save” button to save and exit the Entity View.

Create a New Query in the Database

Queries allow users to save a filtered set of data within the database without selecting the same filters every time. It is extremely useful for showing only certain pieces of information in the database and can be customized for many different views.

1. Under the “Filter” tab on the left of the database, scroll down and click on the “Decision” class.
 - *Note: This now only shows entities that are of the “Decision” class.*
2. Click on the “Saved Queries” tab on the left menu then select, “Save Current Query.”
3. Give the query the following name: “Project Decisions Query.”
4. Click “Save to Project.”

Notes on Queries

Multiple Queries can be saved and are all stored in the “Saved Queries” tab. Once a query has been saved, users can modify the name and settings if needed. This can be done by clicking the wrench or pencil icon where the query is saved. As more entities are added to the project, the database will automatically add entities in their appropriate database query. For example, if another decision entity was added to the database, Innoslate would automatically include the new decision in the query.

Tip: Save queries to the project so that all users can view and access the saved query. Just clicking “save” will only save the query for that specific user.

Lab 3. Develop Scenario Model

Applying LML to the context diagram emphasizes using Input/Output (I/O) and Asset classes, combining functional and physical elements. There are two options in Innoslate for implementing a context diagram:

- 1) The Asset Diagram - Treat the Conduits similarly to Input/Output class entities.
- 2) The Physical I/O Diagram - Show the I/O flows between Assets.

Given that the goal of functional analysis and allocation is to define the functional elements, the data flowing between them, and the allocation to the Assets and Conduits; the Physical I/O Diagram is the preferred approach.

The entity mapping below describes the internal relationships that the Physical I/O covers. The purpose of this mapping diagram is to illustrate how Innoslate organizes the internal relationships between elements and makes the creation of the Physical I/O a more intuitive experience.

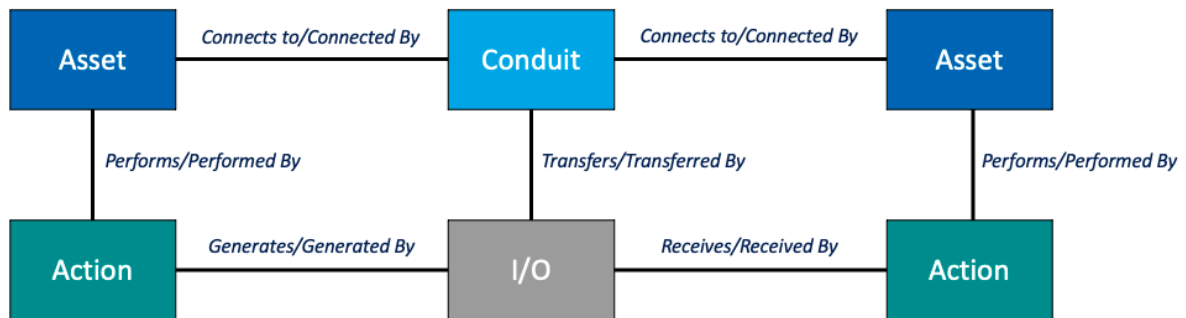


Figure 17. Physical I/O Entity Mappings

Create the Scenario Mode Physical I/O

Now that there is a general understanding of how the Physical I/O manages its entity interrelationships, proceed to create a Physical I/O diagram.

1. Go to “Diagrams View” from the top Menu Bar.
2. Select “New Diagram” then “Physical I/O” from the drop-down list provided.
3. Fill in the following Attributes:
 - a) Number: “OCD.1”
 - b) Name: “Operational Context Diagram – Phase 1”
 - c) Description: *Provide a description for this diagram*
4. Follow the entity names, numbers, and descriptions as described in the figure below.
 - Note: Your goal will be to match the diagram that’s shown below in Figure 18.

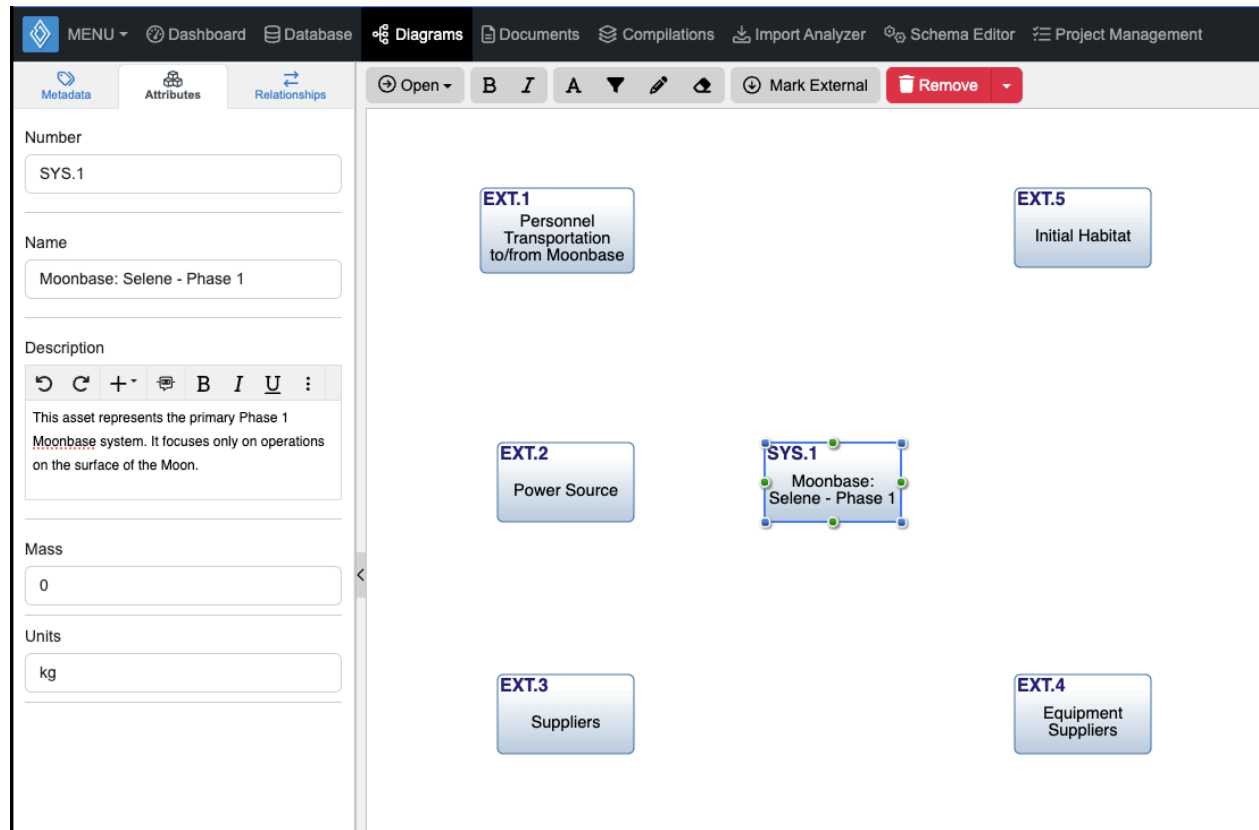


Figure 18. Beginning Physical I/O Diagram

5. For organization, assign the “External System” label to the “EXT” entities by going to their Metadata tab on the left side menu bar.
 - *Note: By connecting the external systems to the original system, you will be asked to create conduits (the connections between systems). Directions and shapes for the conduits can be changed by selecting the line and using the “Line Options” button at the top.*
6. Follow the table below to create the completed Physical I/O diagram, as shown in Figure 19.

Table 2. Physical I/O Entity Attributions

Number	Name	Conduit	Action To	Action From	Origin
SYS.1	Moonbase: Selene - Phase 1	Personnel Returning	Prepare Personnel for Return to Earth	Receive Personnel	SYS.1 to EXT.1
EXT.1	Personnel Transportation to/from Moonbase	Personnel for Moonbase	Send Personnel	Receive Personnel	EXT.1 to SYS.1
EXT.2	Power Source	Power Source	Provide Power	Receive Power	EXT.2 to SYS.1
EXT.3	Suppliers	Suppliers	Provide Supplies	Receive Supplies	EXT.3 to SYS.1
EXT.4	Equipment Suppliers	Equipment	Send Equipment	Receive Equipment	EXT.4 to SYS.1
EXT.5	Initial Habitat	Habitat with Transporter	Provide Habitat Resources	Receive Habitat Resources	EXT.5 to SYS.1

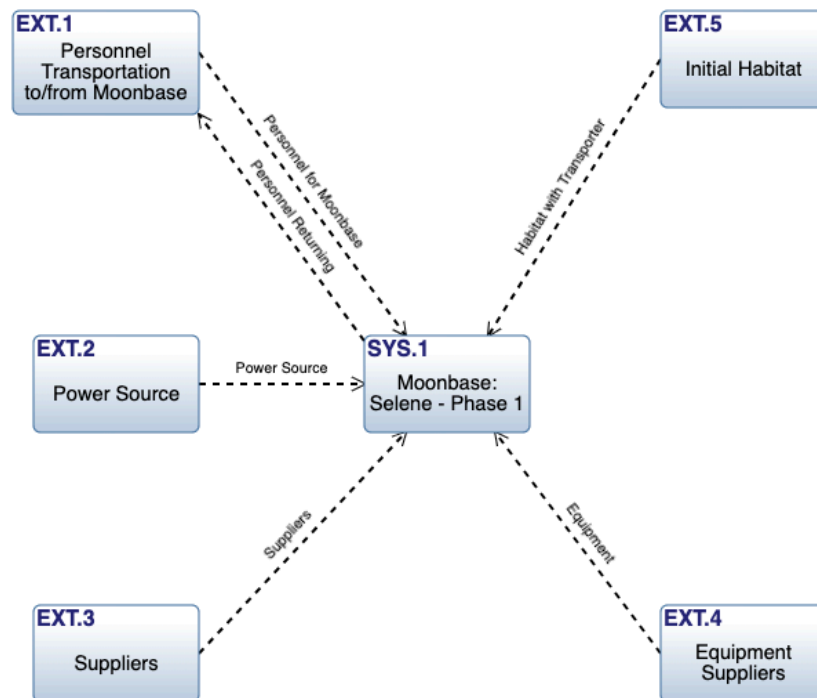


Figure 19. Completed Physical I/O Diagram

New Input/Output Creation

No 'Input/Output' entity selected Create

Personnel Returning (Conduit)

Prepare Personnel for Return to Earth (Action)

Receive Personnel (Action)

Select Directionality

☒ From SYS.1 Moonbase: Selene - Phase 1 to EXT.1 Personnel Transportation to/from Moonbase

☐ From EXT.1 Personnel Transportation to/from Moonbase to SYS.1 Moonbase: Selene - Phase 1

Select Origin

☒ Conduit having no directionality

☐ Conduit going from SYS.1 Moonbase: Selene - Phase 1 to EXT.1 Personnel Transportation to/from Moonbase

☐ Conduit going from EXT.1 Personnel Transportation to/from Moonbase to SYS.1 Moonbase: Selene - Phase 1

☐ Conduit being bidirectional

Cancel Create

Figure 20: Creating Related Entities for the Input/Output Entity

Users must create new Conduit or select an existing one for each part of the I/O Creation. Pay careful attention to the direction of the inputs and outputs and which Actions will “Provide” or “Receive” an Asset. Once the I/O is created, change the Line Option to Straight and name it, “Personnel Returning.”

Create an action diagram with I/O

As a result of building this diagram, the outcome includes 12 Actions, 5 Conduits, and 6 I/O entities. Six of these Actions are performed by the system. This provides a foundational start for the functional system model. These actions can be utilized to create a top-level action diagram that receives inputs and outputs, serving as the functionality for the moonbase.

1. Go to “Diagrams View” from the top Menu Bar.
2. Select “New Diagram” then “Action Diagram” from the drop-down list provided.
3. Fill in the following attributes:
 - a) Number: “M”
 - b) Name: “Top-Level Moonbase Functionality”
 - c) Description: “This action contains the top-level of Moonbase functionality, which will be used to derive requirements for the overall system.”
4. On the left menu bar, enter the “Existing” Tab.
5. Start typing in the Action Entities from the pull-down menu as shown in Figure 19 below.

- *Note: The entities below have been automatically created in the previous step of creating the Physical I/Os. When creating elements for “Action to” and “Action from” for each asset and its conduits, the system automatically allocates new action entities that can be recalled in a new action diagram.*
6. Renumber the action entities and manually create the Input/Output elements to replicate Figure 21 below.

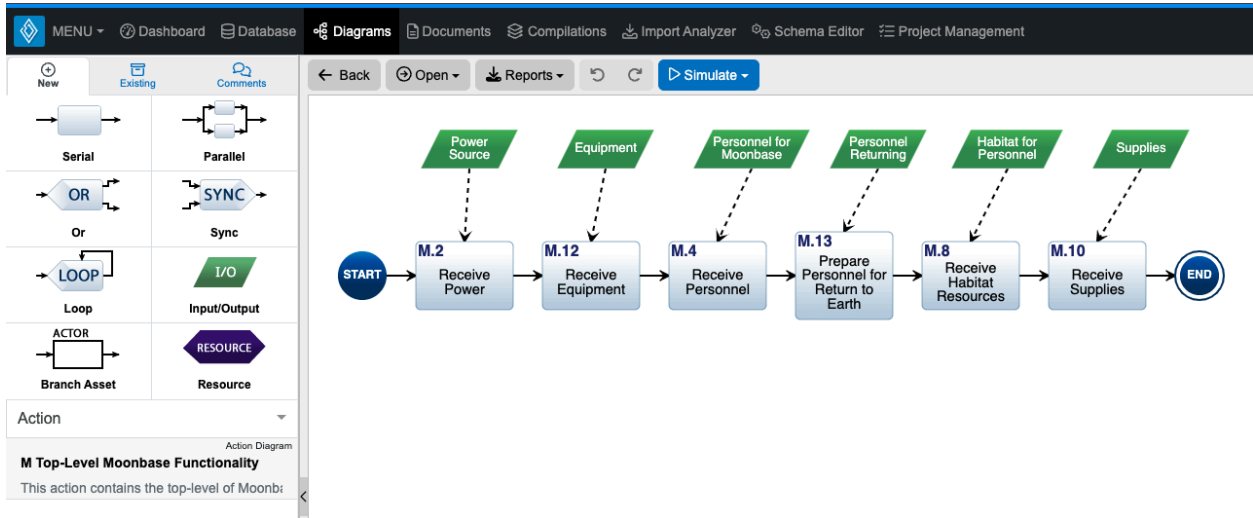


Figure 21. Top-Level Action Diagram

Lab 4. Execute Scenario Model

Create an Action Diagram to Represent Different Scenarios

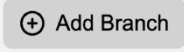
1. Navigate to “Diagrams View” from the top menu bar
2. Click on “Create Diagram” and select “Action Diagram.”
3. Fill in the following attributes:
 - a) Number: “S.”
 - b) Name: “Construct and Operate the Moonbase”
 - c) Description: “This action contains the top-level of Moonbase functionality, which will be used to derive requirements for the overall system.”
4. Click “Finish” to create the diagram.
5. Drag a “Sync” entity and drop it on the line.
6. Click on the middle of the parallel arrows to highlight the entity's base.
 - a) *Note: Clicking the middle where the arrows meet should highlight a rectangle. Highlight over the area with the mouse for the rectangle to appear, then click on it.*
7. Once highlighted, click the  button to add a new branch.
8. Keep adding until there are 8 total branches.
 - a) *Note: This will make the diagram large, but the size can be adjusted once entities are on the branch.*
9. Drag “Serial” entities onto each branch and fill in attributes using the following table:

Table 3. Scenarios with Descriptions

Scenario	Description
S.1 Conduct Initial Site Survey	Initial site survey is conducted robotically. Its primary purpose is to understand the full information needed to choose specific sites for habitats (short and long term), landing/takeoff “pads,” excavation, and production of oxygen.
S.2 Initial Human Landing and Site Setup	The first human landing will establish the base camp using lunar lander and begin preparation for receiving supplies and equipment. <i>Note: some will be staged ahead of time.</i>
S.3 Receive Supplies and Equipment	Receive first shipments of supplies, power source, and equipment.
S.4 Build New Habitat	Receive shipped habitat and build new habitat using materials mined from Moon for shielding (at least).
S.5 Receive New Personnel	Receive new personnel for assignments. This includes training of new personnel for specific assignments.
S.6 Rotate Personnel Out	Prepare personnel to return to Earth. This includes preparation of the ship for return (refueling, restocking, any maintenance needed).
S.7 Construct Oxygen Extraction System	Build facilities to extract the oxygen from the regolith.

S.8 Extract Oxygen from Regolith	Conduct operations to extract oxygen.
S.9 Construct and Operate Moonbase	Complete full scale construction of the Moonbase and operations to support Phase 1 goals. This scenario becomes the integrated behavior model for Phase 1.

10. Click on the “Existing” tab located on the tab to the left.
11. In the search bar, select the following Input/Output class entities:
 - a) “Equipment”
 - b) “Supplies”
 - c) “Power Source”
 - d) “Habitat with Transporter”
 - e) “Personnel for Moonbase”
 - f) “Personnel Returning”
12. Drag the entities onto the diagram to match Figure 22.
 - o *Note: When dragging, remember to match the arrow direction and make each entity a “Trigger” relation. S.1, S.3, and S.4 are not decomposed yet; that will happen later in this lab.*
13. Recreate S.9 and S.10 entities to complete the “S Construct and Operate Moonbase” action diagram.

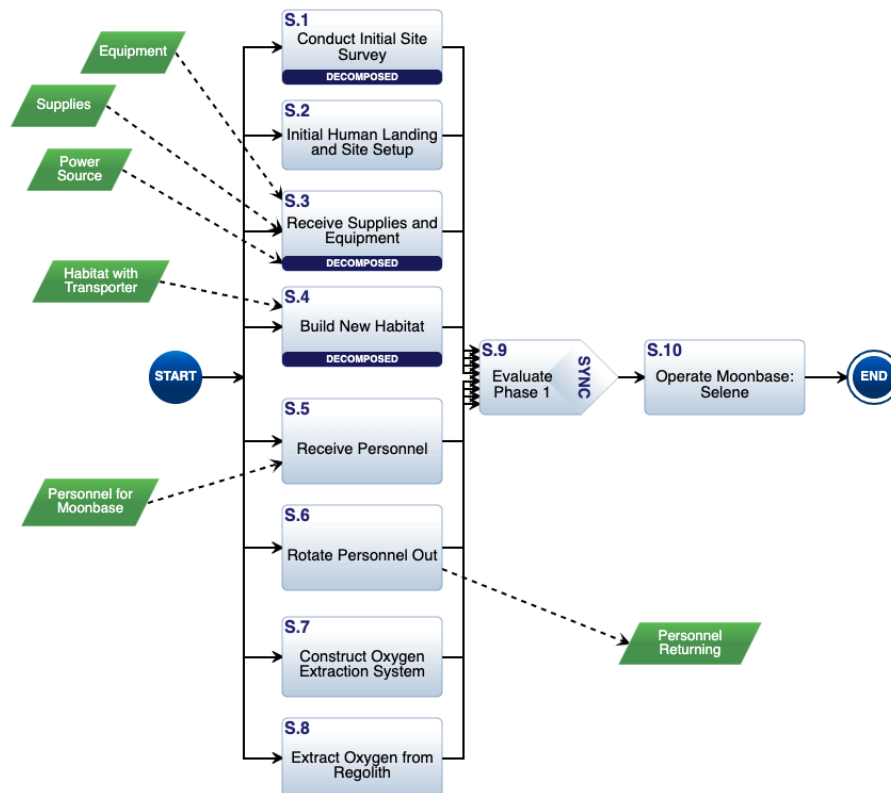


Figure 22. Scenarios for Constructing and Operating the Moonbase

Note: The Parallel Entity executes all the paralleled Action Blocks. The fastest block to complete its task will negate the rest of the Action Items that are in parallel and will move on to the next sequenced Action item. This Parallel Action can represent the transition to full-scale operations.

Decompose a Scenario

1. Click on the entity, “S.1 Conduct Initial Site Survey,” then click the “Open” dropdown and select “Decomposition Diagram.”
2. Build the Action Diagram that’s shown below in Figure 23.

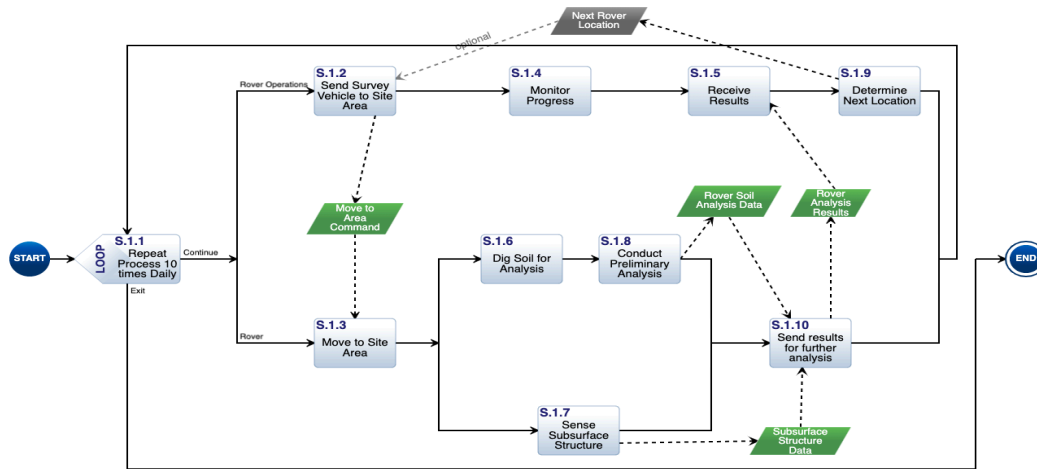
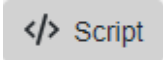


Figure 23. Decomposed Action Diagram for Conduct Initial Site Survey

3. Select the entity, “S.1.1 Repeat Process 10 times Daily,” then click  near the top menu bar.
4. Keep “Loop Iterations” selected, then type “10” for iterations.
5. Push the blue “Submit” button.
6. Shift-Select the bottom 3 entities in parallel lines (Dig Soil for Analysis, Conduct Preliminary Analysis, and Sense Subsurface Structure), as well as the rectangles towards the middle of the parallel lines.
 - o *Note: When all are selected, the outline should look like the figure below.*

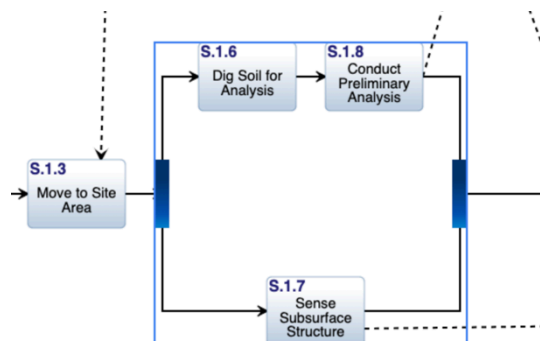


Figure 24: Highlight of Decomposition

7. Click on the  Decompose button near the top toolbar.
8. Build the decomposed Action Diagram for the Survey Area, similar to Figure 25.
 - o The purple hexagon is a Resource entity. Resources entities can be consumed, produced, or seized by an Action.

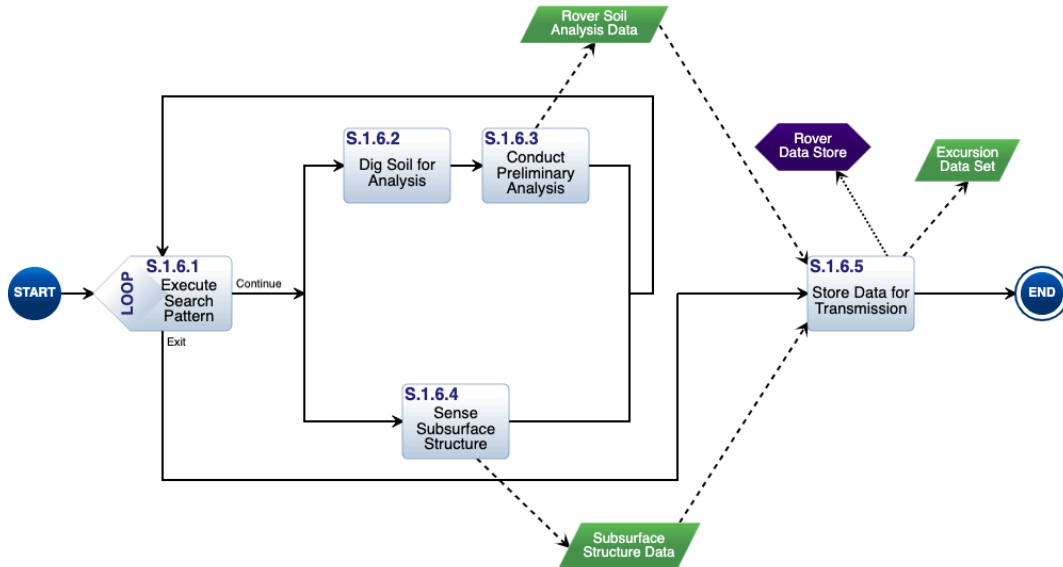


Figure 25: Survey Area Decomposition

9. Update the Action Diagram for “Conduct Initial Site Survey” to match Figure 26.
 - o Note: This will also include updating the Input/Output entities (I/O's) to match.
 - o Tip: Use existing I/O's when possible, and create only what you need to keep from duplicating entities.

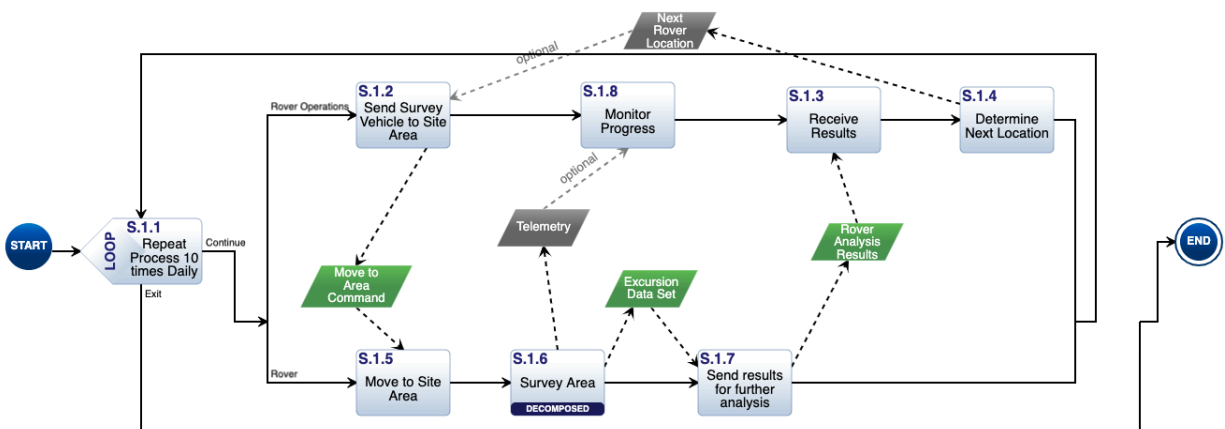


Figure 26. Updated Scenario 1 Model

10. Drag the entity, “Branch Asset (Actor),” to a line to assign the Asset that will perform the specified actions.
 - o *Note: They can be placed on the parallel branches of the actions. In this case, it will be “Rover Operations” and “Rover.”*

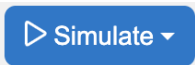
Performing Discrete Events Simulation

Now that an action diagram has been created to represent the structural model, it can be simulated to show real-time data.

1. While in the decomposed diagram, go through each of the listed entities in the following table and assess the durations to impact the simulation:

Table 4. Duration Attributes for Action Entities

Entity	Duration
S.1.1 Repeat Process 10 Times Daily	0 seconds
S.1.2 Send Survey Vehicle to Site Area	triang.dist(0.5, 1, 0.75) hours
S.1.3 Receive Results	triang.dist(0.5, 1, 0.75) hours
S.1.4 Determine Next Location	triang.dist(1, 2, 1.5) hours
S.1.5 Move to Site Area	triang.dist(0.5, 2, 1) hours
S.1.6 Survey Area	0 seconds
S.1.6.1 Execute Search Pattern	0 seconds
S.1.6.2 Dig Soil for Analysis	triang.dist(1, 5, 3) minutes
S.1.6.3 Conduct Preliminary Analysis	triang.dist(4, 10, 6) minutes
S.1.6.4 Sense Subsurface Structure	triang.dist(5, 10, 7) minutes
S.1.6.5 Store Data for Transmission	triang.dist(0.5, 2, 1) minutes
S.1.7 Send Results for Further Analysis	triang.dist(5, 20, 15) minutes

2. Once the entities are assigned their durations, click the  button and choose “Discrete Event” from the drop-down.

- o *Note: The Discrete Event simulator has various widgets that allow users to view their desired data. The left sidebar is where users can find categories such as cost, time, resources, and other widgets that they can select and add to their interface.*
3. Explore the different categories and choose the following widgets to enable on your interface:
 - a) "Status"
 - b) "Action Trace 3D"
 - c) "Tree Map"
 - d) "Console"
 - e) "Total Time"
 - f) "Total Cost"
 - g) "Cost Over Time (Total)"
 4. Organize your Discrete Event Simulation interface to look like the following image:
 - o *Tip: Adjust the size of each widget by dragging the corner of the box.*

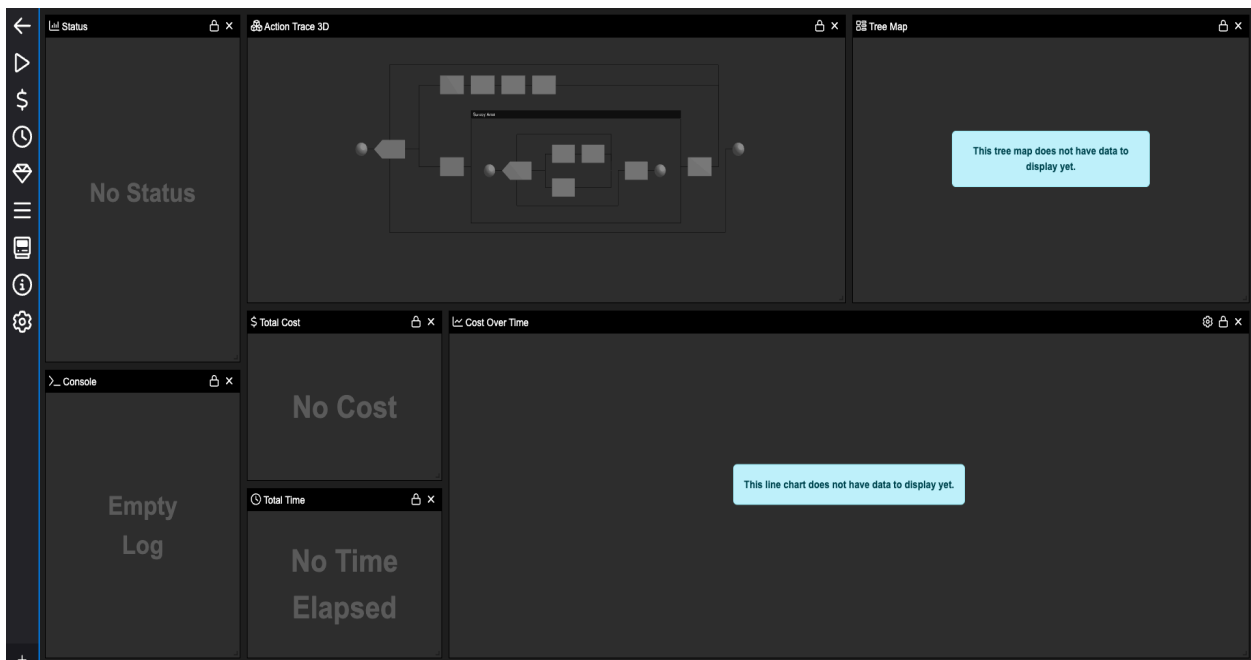


Figure 27. Organized Simulator Interface

5. Once the interface setup has been finalized, click the  (settings) icon in the left sidebar and apply the following settings:
 - a) Display Mode: "Dark"
 - b) Calendar Mode: "True"
 - c) Speed: "10000x"
 - d) Decisions: "Prompt on no script"
 - e) Start: "Normal"

6. Click “Save Changes.”
7. Finally, click the  (start) button to simulate your action diagram and gain results based on all of the information in each entity.

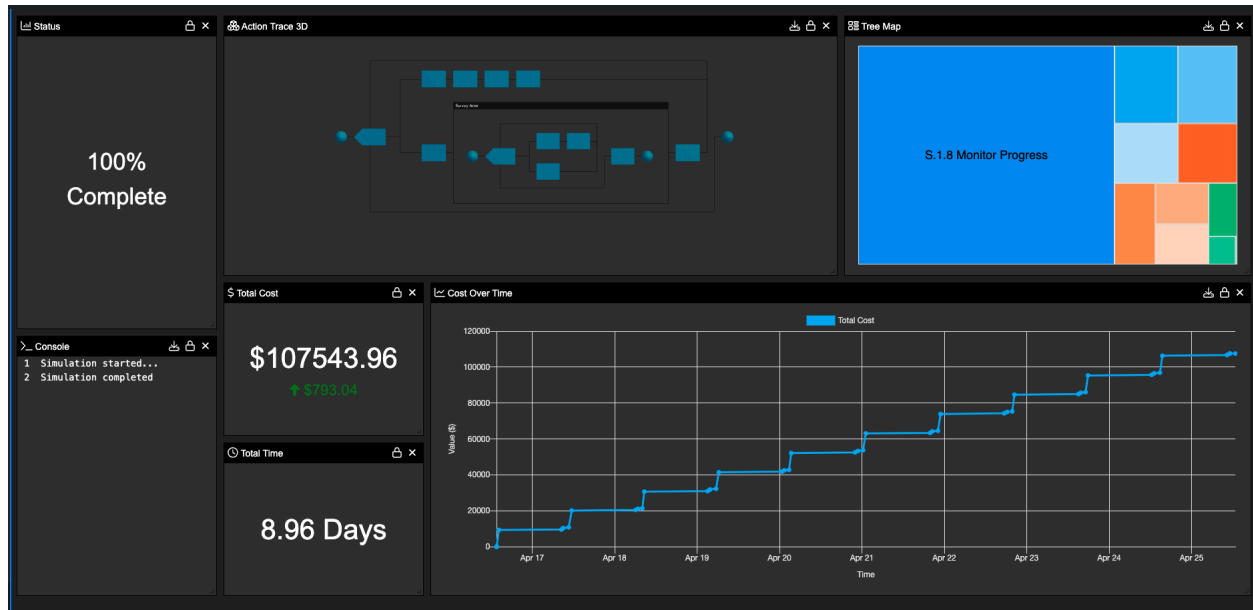


Figure 28. Simulated Simulator Interface Results

- o Note: Users can toggle the simulation speed, pause the simulation, or reset the simulation using the features available in the left sidebar during the simulation.

Lab 5. Develop CONOPS Scenario Matrix

Create a Concept of Operations (ConOps) Document from the Documents view.

1. From the “Documents” tab on the top toolbar, click on .
2. In the “Type” dropdown field, select “Concept of Operations Document.”
3. Under the “Specify” dialog box, add any desired **number**, **name**, and **description**, then select “Next.”
4. Under the “Template” dialog box, click on the dropdown menu and select “ASSE Concept of Operations (with guideline).”
 - o *Note: This is the American Society for Safety Engineers template for ConOps.*

Section 5 is where you can start recording the scenarios for Moonbase.

1. Scroll down and select “section 5” to start editing that section.
 - o *Note: Section 5 can be found by scrolling on the document or through the Hierarchy sidebar located on the right.*
2. Once selected, edit the empty “Operational Scenarios” table to reflect the scenarios listed in Table 2.

Table 5. Initial List of Scenarios for Moonbase: Selene - Phase 1

Scenario	Description
1. Conduct Initial Site Survey	Initial site survey is conducted robotically, its primary purpose to understand the full information needed to choose specific sites for habitats (short and long term), landing/takeoff “pads,” excavation, and production of oxygen.
2. Initial Human Landing and Site Setup	The first human landing will establish the base camp using lunar lander and begin preparation for receiving supplies and equipment. (<i>Note: Some will be staged ahead of time</i>)
3. Receive Supplies and Equipment	Receive first shipments of supplies, power source, and equipment.
4. Build New Habitat	Receive shipped habitat and build new habitat using materials mined from Moon for shielding (at least).
5. Receive New Personnel	Receive new personnel for assignments. Includes training new personnel for specific assignments.
6. Rotate Personnel Out	Prepare personnel to return to Earth. Includes preparation of ship for return (refueling, restocking, any maintenance needed).
7. Construct Oxygen Extraction System	Build facilities to extract the oxygen from the regolith.
8. Extract Oxygen from Regolith	Conduct operations to extract oxygen.
9. Construct and Operate Moonbase	Perform full scale construction of the Moonbase and operations to support Phase 1 goals. This scenario becomes the integrated behavior model for Phase 1.

3. Select section 5.1 (located directly below section 5) and edit the empty “Scenario Characteristics and Descriptions” table to reflect Table 6.

Table 6. Scenario Characteristics for Moonbase: Selene - Phase 1

Characteristics	Description
Environment	Moon Normal (average radiation levels), Solar Flare
No. of Daily Excursions	Trips taken by robotic or human assets: 0, 1, 5, 10
No. of Personnel at Site	0, 5, 10, 50, 100
No. of Monthly Landings	Landing of rockets carrying people, supplies, and/or equipment: 0, 1, 5
Amount of O2 Discovered per Excursion (tons)	0, 10, 100, 1000, 10000
Amount of O2 Extracted (tons/month)	0, 50, 150

- Select section 5.2 (located directly below section 5.1) and edit the “Applying the Characteristics to Operational Scenarios” table to reflect Table 7.

Table 7. Scenario Matrix for Moonbase: Selene - Phase 1

Scenario	Characteristics					
	Environment	Daily Excursions	Personnel at Site	Monthly Landings	O2 Found	O2 Extracted
1.	Normal	10	0	0	10	0
2.	Normal	5	5	0	0	0
3.	Normal	1	5	0	0	0
4.	Solar Flares	5	5	0	0	0
5.	Normal	5	10	1	0	0
6.	Normal	1	10	1	0	0
7.	Normal	5	50	5	100	0
8.	Solar Flares	10	50	5	1,000	50
9.	Solar Flares	10	100	5	10,000	150

Add the Characteristics’ Attributes in the Schema Editor.

- Open the “Schema Editor” (located on the top toolbar).
- On the left menu bar, go to “Classes” and select  called, “Scenario Characteristic.”
- Make the Parent Class “Characteristic.”
- Add the Attributes that are listed under the Characteristics header from Table 4:
 - Under the “Type” field for each attribute, choose “MULTISELECT” and fill in the available options that are listed from each section in Table 3 above.
 - For the “Default” field, choose “Moon Normal” and “0” for the rest of the numerical values so the ending table looks like Figure 29.
 - Tip: Make minimal edits to the Schema Editor, as it can be confusing for your team.*

Attribute Name	Type	Order	Choices	Default	
Value	TEXT	1			 
Environment	MULTISELECT	3	Moon Normal, Solar Flare	Moon Normal	 
Daily Excursions	MULTISELECT	4	0, 1, 5, 10	0	 
No. of Personnel at Site	MULTISELECT	5	0, 5, 10, 50, 100	0	 
No. of Monthly Landings	MULTISELECT	6	0, 1, 5	0	 
Amount of O2 Discovered per Excursion (tons)	MULTISELECT	7	0, 10, 100, 1000, 10000	0	 
Amount of O2 Extracted (tons / month)	MULTISELECT	8	0, 50, 150	0	 
Units	TEXT	10			 

Figure 29. Characteristics Additions in Schema Editor

Create a Hierarchy Diagram for the Moonbase Scenarios.

- Go to “Diagrams View” and click . Select “Hierarchy Diagram” with the root entity being an Action class.
- Specify the diagram with the following attributes:
 - Number: “SC”
 - Name: “Scenario Characteristics”
 - Description: *Give a brief description of the diagram*
- On the blank palette, build the scenario hierarchy as it appears in the diagram in Figure 30.

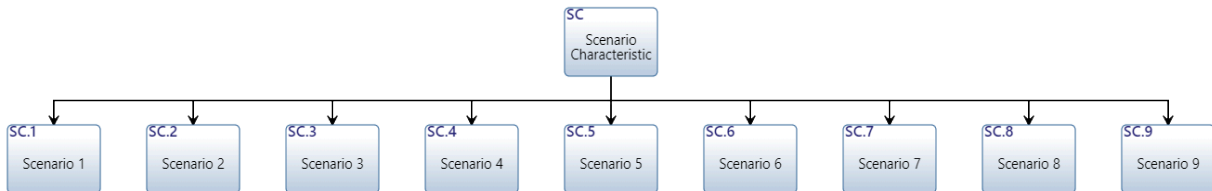


Figure 30. Hierarchy Diagram for Moonbase Scenarios

- In each scenario entity, fill out the description attribute as they are listed in Table 2 above until all 9 scenarios are filled in.
 - Note: Always remember to click save the diagram when finished.*

Create a Relationship between both Diagrams

- Open the Diagram “S. Construct and Operate the Moonbase.”
- Select the top Action, “S.1 Conduct Initial Site Survey.”
- On the left menu, go to the “Relationships” tab. Click on the dropdown menu that currently says “Pinned” and select “All.”

4. Scroll down to the bottom where it says, “*specified by* Characteristic.” Click on the arrow next to the  button and select “Relate Existing Entities.”
5. Type the number from Scenario 1 (SC.1) in the search bar, then select the available entity to relate and create the relationship.
6. Repeat the process of Step 5 for all 9 scenarios in the parallel diagram.
 - o *Note: Be sure to match the scenario character number to the corresponding action number.*

Lab 6. Identify Potential Implementations

Using components that have been identified from previous chapters and additional research, we begin to understand that our Assets in this project may have similar Assets that would be convenient to have a field for values to be entered in.

To capture the relevant information about the Rover (or any other component of the system), we would use the LML Asset class entities. Since the general Asset class does not have many attributes, we have the option of using Characteristics or adding attributes to the Asset class. We usually use Characteristics when we are creating a common element that we want to reuse. An example of this would be painting equipment a certain color, such as Battleship Gray¹. Battleship gray has a specific set of color values. We would want to make sure we used the same exact colors everywhere it is required. A Characteristic could then be created with this information and then related to any appropriate Asset.

Add attributes to a class using the Schema Editor

1. Navigate to the “Schema Editor.”
2. Open up the “Classes” tab. Find the “Asset” class and click into it.
3. Click the blue “Add Attributes” button on the right. Follow the figure below to add the “Mass” and “Units” attributes.

The screenshot shows the 'Schema Editor' interface. On the left, there's a sidebar with a pie chart and metadata for the 'Asset' class. The main area is titled 'Edit Asset Class' and contains a form with the following fields:

- Name: Asset
- Description: An Asset entity specifies an object, person, or organization that performs Actions, such as a system, subsystem, component, or element.
- Hidden: No
- Parent: Select option
- Abstract: No

Below the form is the 'Attributes' section, which contains a table:

Attribute Name	Type	Order	Choices	Default
Mass	NUMBER	1		
Units	TEXT	2		kg

An 'Add Attribute' button is located in the top right corner of the Attributes section.

Figure 31. Adding Mass and Units Attributes to the Asset class entities

Create Assets with Newly Defined Attributes

Note: We will be using the Rover as defined in Chapter 5 of the Real MBSE book to create the attributes that we just defined.

1. Go to the “Database” tab at the top menu bar.
2. Push the  button and select “Asset.”

- o *Note: This page is called the “Entity View.” It contains all of the information pertaining to that entity. The “Global ID,” “Class,” and “Labels” sections are on the left side. This information may be useful as you become more comfortable using existing entities in Innoslate.*
- 3. In the “Metadata” tab on the left menu bar, scroll down and select the label, “Subsystem,” for the Rover.
- 4. Enter the following information:
 - a) Number: “ROV.1”
 - b) Name: “Rover”
 - c) Description: “Mass includes primary instruments:
 - Near Infrared Spectrometer
 - Neutron Spectrometer
 - Drill
 - Oxygen & Volatile Extraction Node
 - Lunar Advanced Volatile Analysis
 - Isotope Measurement of Volatile
 And other instrument candidates:
 - Radiation monitor
 - Seismometer
 - Heat flow measurement
 - Spectro-microscope camera
 - Active x-ray spectrometer”
 - d) Mass: “349”
 - e) Units: “kg”
- 5. Add a picture to the “Metadata” tab:
 - a) In the “Metadata” tab on the left menu bar, click on the upload button () above the “Global ID/ID/Class” section and assign the desired picture.
 - o *Note: You can find a picture of a rover by googling “moon rover.” Just make sure to source your picture!*
- 6. Establish a relationship:
 - a) Locate the “Relationships” section to the right of the screen and scroll down to “performs Action.”
 - b) Click on the arrow next to the  button then select “Relate Existing Entities.”
 - c) In the search bar, type the entity, “Move to Site Area” (Number S.1.5). Select the available entity to relate and create the relationship.
 - d) Repeat the previous step to relate entity “Send Results for Further Analysis (Number S.1.7).
 - o *Note: Make sure to click the “Save” button on the top. The finished product should look similar to Figure 32.*

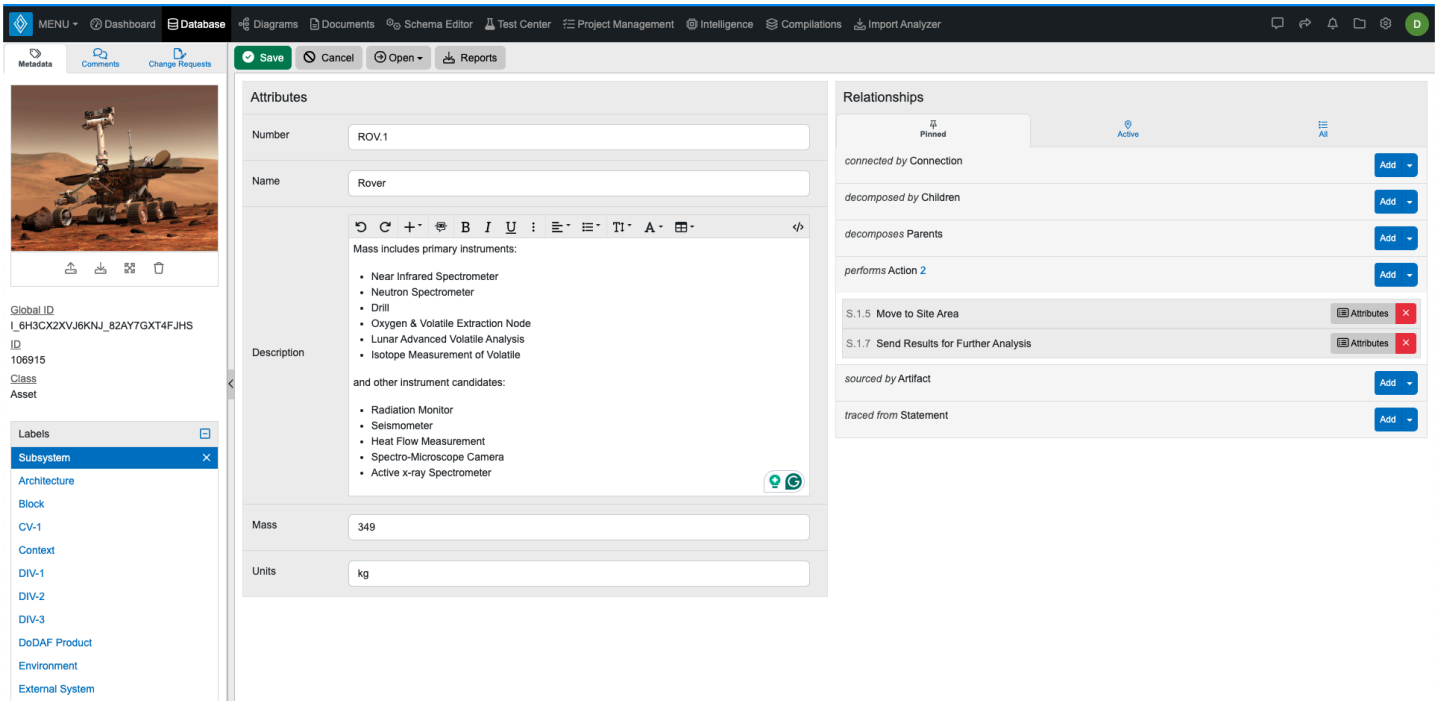


Figure 32. Implementing the Rover subsystem

Create a Label in the Schema Editor

- o *Note: Creating and adding labels to different entities will keep Rover possibilities organized within the project.*
- 1. Navigate to the “Schema Editor.”
- 2. Go to the “Labels” tab and push the  button.
- 3. Name the new label, “Rover.”
- 4. Provide the description, “Used to clarify possible rover options.”
- 5. Assign the “Class(es)” to “Asset” and save.

Add and Organize New Rover Labels in the Database

Add one more rover to the database to showcase created labels. For organizational purposes, make sure to go back to the ROV.1 Rover entity in the Database. Next, add one more Rover to the Moonbase project for completeness.

1. Go to the “Database” and create a new “Asset” entity.
2. **Number** the Asset, “ROV.2,” and **Name** it, “ATHLETE Rover.”
 - o *Note: The concept for the ATHLETE Rover is outlined in Chapter 6 of the book.*
3. Copy the following text into the **Description** box:

JPL leads the development of the ATHLETE (All-Terrain Hex-Legged Extra-Terrestrial Explorer) robotic vehicle. Other team members include NASA’s John and Ames Centers, Stanford University, and Boeing. ATHLETE is capable of rolling over undulating moon-like terrain and

“walking” over extremely rough or steep terrain so that the rough surface of the moon can load, transport, manipulate, and deposit payloads.

The first version of the ATHLETE vehicle is the ability to dock with special-purpose device, including a launchable / releasable grappling hook, refueling stations, excavation implements. Development plans for future versions include the ability to move 100 times faster than the Mars Exploration Rovers and traverse almost any terrain, including vertical rock faces or sandy slopes at the angle of repose by using a launchable / releasable grappling hook.

Date Accessed: [Today’s Date]

Source: https://www.nasa.gov/multimedia/imagegallery/image_feature_748.html

4. The **Mass** of the ATHLETE is 2340 kg.
5. Add the picture from the NASA website to the “Metadata” tab.
 - *Note: A picture is provided at the NASA website above; remember to source this information. Upload it as the picture for this entity. Instructions to do this are found earlier in this lab.*
6. Add both the “Rover” and “Subsystem” labels.
7. Lastly, add the “performs” relationship to the following actions:
 - a. S.1.5 Move to Site Area
 - b. S.1.7 Send Results for Further Analysis
 - *Note: Instructions on how to use the “Relate Existing Entities” function can be seen in the process earlier in the lab (Step 6, Establish a Relationship).*

Lab 7. Identify Trade Studies Needed

“A systems engineer doesn’t get paid for doing trades, she gets paid for deciding which trades to do.” – Sandra Cauffman

Trade studies are in the Schema Editor under the Artifact class. We will need to add some trade studies to the project to make sure that we can allocate specified trade studies to the correct Requirement or Statement.

Add Trade Study Artifacts to the Database

1. Go to the “Database View” and click on “New Entity,” then add a new “Artifact.”
2. Fill out the attribute fields with the following information:
 - a) Number: “TS.1”
 - b) Name: “Construction Materials from Moon Trade Study”
 - c) Description: *Leave this blank for now*
 - d) Date Published: *Current Date*
 - e) Label: “Trade Study”
 - f) File: *N/A*
 - o *Note: We can upload any file that may pertain to this Trade Study using the Upload button in this field. This is convenient for file keeping so that everything is in one place.*
3. Push the green “Save” button at the top of the menu bar.
4. Repeat the process from Step 1 to create the following trade studies in Tables 8-10

Table 8. Trade study 2

Trade Study 2	
Number	TS.2
Name	Oxygen Levels Required from Moon Trade Study
Description	This trade study determines the range of oxygen needed to support life for the first phase of Moonbase operations. It sets the limit for meeting the verification requirements VR.1.3.1. It should also estimate levels needed for future phases of the program.
Date Published	3/7/2024 12:48:52 PM
Label	Trade Study

Table 9. Trade study 3

Trade Study 3	
Number	TS.3
Name	Oxygen Need for Rocket Fuel
Description	This trade study determines the amount of oxygen needed for rocket fuel by the end of Phase 1 of Moonbase construction. It should also estimate needs for future phases of the program.
Date Published	3/7/2024 9:40:53 AM
Label	Trade Study

Table 10. Trade study 4

Trade Study 4	
Number	TS.4
Name	Robotic vs Human Survey and O2 Extraction
Description	The purpose of this trade study is to determine how much human oversight of the robotic survey and O2 extraction capabilities is required.
Date Published	TBD
Label	Trade Study

Insert the Trade Studies into the Project

- Open "R. Top-Level Requirements Document for Selene."
 - Note: We will need to add more requirements for robotic or human landings. They will be vague to begin with and may be subject to further validation from a requirements team.*
- Under the 1.3 Statement, add a Child statement with the following attributes (the numbers will populate automatically as 1.3.2):
 - Name: "Land robotic survey rover"
 - Description: "The system shall land a robotic survey rover."
 - Rationale: "The robotic survey is required for the initial surveying to identify where and how much O2 can be used to produce O2 for humans and as rocket fuel."
- Push the  button to add another child statement to 1.3.
 - Note: If you select, "Add Child," it will be a child statement to clarify the 1.3.2 statement; 1.3.3 is intended to go here.*
- Fill in the following attributes:
 - Name: "Land astronauts to support survey and extraction O2."

- b) Description: “The system shall land astronauts to support the surveying and extraction of O2.”
- c) Rationale: “Humans will be needed to conduct detailed surveys and build systems to extract oxygen.”

Connect the Requirements to the Trade Study

Note: This will be done by selecting each requirement and using the relationships tab. Add the trade study created earlier to these requirements, as shown in Figure # for the second new requirement (1.3.3).

1. In the “R. Top-Level Requirements Document for Selene” document, select requirement 1.3.3 and click on the “Relationships” tab on the left side menu.
2. From the “Pinned” drop-down menu, scroll down to “traced to Many.” Click on the arrow next to the  button, then select “Relate Existing Entities.”
 - o *Note: Details for the All / Pinned / Popular menu selections will be at the bottom of this section.*
3. Push “Enter” to add the relationship, then click the green check mark button. You will see the Name of the trade study that’s now allocated to the Statement in your Requirement Document.

Notes for the Relationships tab:

- o The “All” drop-down menu lists every relationship that exists in LML among the entities. You can find any relationship you need in this menu.
- o The “Popular” drop-down menu is a compilation of the most used relationships in your project. This menu might change over time as more users come onto your project.
- o The “Pinned” drop-down menu is a personalized menu of preferred relationships and is specific to only that user. Instructions on how to pin/unpin relationships are described below:
 - I. In the “Relationships” tab, click on the drop-down menu and select “All.”
 - II. Highlight over any relationship and click on the  button next to the relationship & class name to pin the relationship.
 - III. Click on the drop-down menu and select “Pinned.” You should see the newly added pinned relationship.
 - IV. Click on the same button again to unpin the relationship.

Lab 8. Design Monitoring

The creation of a Computer-Aided Design (CAD) model will review the results from a typical design engineering activity. The mechanical engineers usually create the models of the components, such as a small lunar rover that might be one of the earliest survey tools.

Download and Extract the Lunar Rover CAD File

1. Click on the following link to access the CAD file:
<https://grabcad.com/library/rover-with-sampling-and-sorting-system-1>
 - o *Note: You may need to create a student/personal account for the download if you don't have one already. It is free and the files can be used for projects or 3D printing in the future.*
2. Click on the main picture at the top and download the file to your computer. Exit from the image window and click on "Download files."
 - o *Note: This will give you the entire database for the lunar rover. Make sure to extract all files from the zipped folder. There should be a "models" folder and an "stl" folder. These will be the files that will be used during this lab.*

Create an Asset Hierarchy of the Lunar Rover

1. Navigate to the "Diagrams" View and click "Create Diagram."
2. Choose a "Hierarchy Diagram" with the root entity class of Asset.
3. Fill out the following fields:
 - a) Number: "AST.ROV.1"
 - o *Note: AST means Asset*
 - b) Name: "NASA Lunar Sampling Rover"
 - c) Description: *Put the CAD website link that contains the files are from and a brief description of the lunar rover.*
4. Build the diagram that you see in Figure 33 to account for all the components of the rover.
 - o *Note: To make the entities stack vertically, go to the wrench on the top right corner of the screen and select "Compress after Level," then choose "1" as the option.*

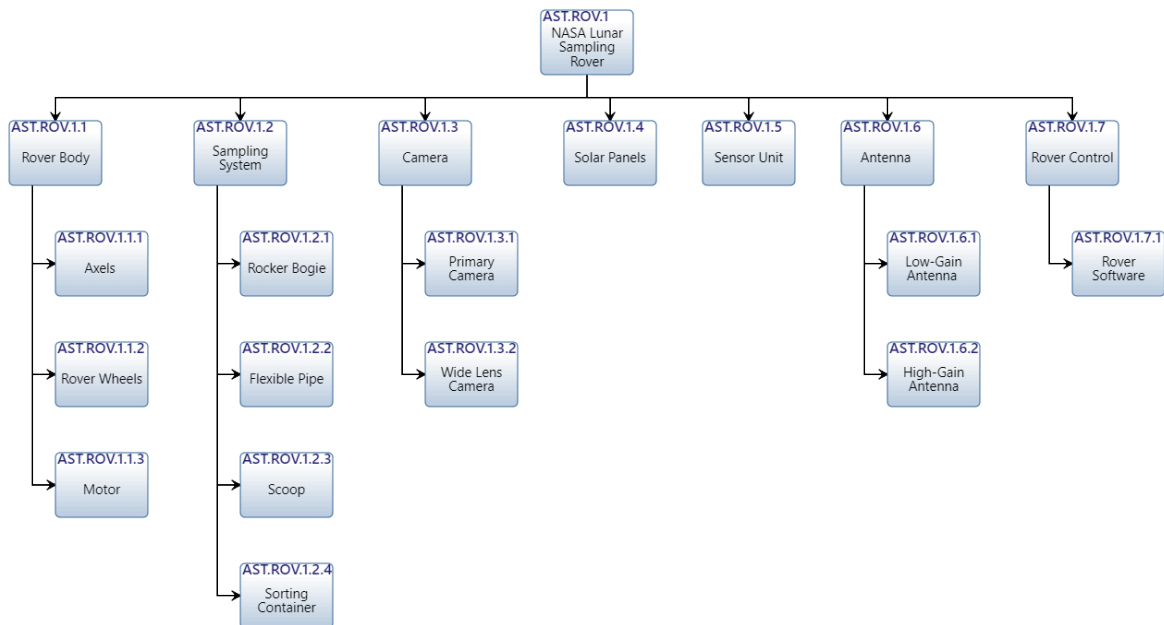
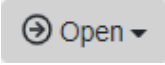
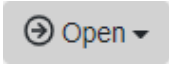


Figure 33: Asset Hierarchy Diagram

5. Select the top-level entity (AST.ROV.1) and navigate to the “Relationships” tab on the left sidebar.
6. Scroll down to “*references Artifact.*” Click the arrow to the right of the  button, then “Create Artifact.”
7. Use the following attributes for the Artifact:
 - a) Number: “ART.Rov.1”
 - b) Name “NASA Lunar Sampling Rover – Artifact”
 - c) Description: *Put the CAD website link that contains the files are from and a brief description of the lunar rover and how it impacts the Moonbase.*
8. Click the green checkmark to save the Artifact.
9. Click the  button on the top toolbar, then select “Entity View.”
10. Click the upload button () on the left sidebar and add the .jpg of the rover.
11. Since the “*referenced by Many*” relationship will be used often in this lab, make sure that it’s pinned as discussed in Lab 7.
 - a) *Note: We will cover the opposite relationship of “references Artifacts” in the next mini-section of the lab.*
12. Push the green “Save” button to return to the hierarchy diagram.

Add both Directions of the “References” Relationship to Pinned

1. Select “AST.ROV.1.1” and click the  button on the top menu bar. Select “Entity View.”
2. To the right on the “Relationships” section, select the “All” tab, then scroll to “references Artifact” and make sure it’s pinned for future use.
3. Select the “Back” button to return to the hierarchy.

During the lab, a similar procedure will be followed for AST.ROV.1.2 and AST.ROV.1.3. This same method will be applied to all other entities listed in the hierarchy (not shown in this lab) to ensure that every part of the vehicle is included in the project.

Create Artifacts for Existing Asset Entities

1. Select AST.ROV.1.1 and go to the “Relationships” tab on the left sidebar.
2. Under the “Pinned” tab, scroll down and locate “references Artifact.”
3. Click the arrow to the right of the  button, then select “Create Artifact.”
4. Use the following attributes:
 - a) Number: “ART.ROV.1.1”
 - b) Name: “Rover Body – Artifact”
 - c) Description: *Leave Blank*
 - d) Date Published: *Current Date and Time*
 - e) File: *Push the upload button and find the “Assem1 – RoverBody – 1.STL” file from the extracted file folder at the beginning of the lab.*
5. Push the green “Save” button on the left side menu.
6. Repeat steps 1-5 for AST.ROV.1.3 using the following attributes:
 - a) Number: “ART.ROV.1.3”
 - b) Name: “Camera – Artifact”
 - c) Description: *Leave Blank*
 - d) Date Published: *Current Date and Time*
 - e) File: *Upload the “Assem1 – camera-2” file*
7. Push the green “Save” button on the top menu bar of the hierarchy diagram to save your work.

Lab 9. Create Test Suites

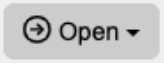
Entity allocations can be complicated, so below are a few key relationships that have been highlighted for your convenience:

- i. **Traced to:** associates Requirements / Statements.
 - o Note: According to the LML Relationships Matrix, it can be used to associate an assortment of classes to Statements; however, it is best to associate Requirements / Statements to an Artifact or other Statements.
- ii. **Verified by:** associates Test Cases / Actions to Requirements.
- iii. **Satisfied by:** associates Performance Requirements to Assets.

“Satisfied by” relationships are defined as the theoretical performance of a system. The “verified by” relationships define the system being tested for verification. In the Moonbase project, test cases can be applied to verify that the functions of the system will perform as expected.

Add a “Verifies” Relationship to the Scoop of the Lunar Rover

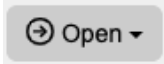
For this lab, verify that the scoop on the lunar rover can successfully be used to collect samples from the Moonbase.

1. Open the “Test Center” from the top Menu bar.
2. A set of Tests for the Rover is provided. Click the existing test suite.
 - o Note: The size of the columns can be adjusted to better see any attribute columns.
3. Select entity “TS.2.4.1.1 Rover Scoop Inspection.”
4. Open its “Entity View” by clicking  , then “Entity View.”
5. Add the following relationships:
 - a) “verifies Requirement,” and add existing entity, “ROV.4 Sample Collection”
 - o Note: Instructions on how to do this can be viewed in Lab 7. Be sure to save all additions made.
 - b) “specified by Characteristic,” and add entity, “STD.57 Scoop Workmanship Standard”
 - o Note: Safety standards such as STD.57 can be defined as characteristics that are structured hierarchically, as demonstrated in Lab 5. Users can develop and update these standards over time as they are learned.
 - c) “incurs Cost,” and add entity, “C.35 Inspector Cost”
 - o Note: Costs can also follow a hierarchical structure and are assigned to specific actions during system development. Users develop these costs over time as they are learned, making C.57 an arbitrary Characteristic number.
 - d) “causes Issues, Risk,” and add entity, “I.257 Scoop Production Problems”

- o *Note: In Figure 29, notice that the scoop inspection failed, which causes the “I.257 Scoop Production Problems” issue. An issue was used instead of risk since it has already occurred. Issues can be added as they are experienced throughout verification. This issue is also an arbitrary issue that has been assigned for learning purposes.*

Add a Traceability Matrix

Traceability Matrices help visualize the relationship between test cases and entities by mapping which test cases are being verified by specific entities (such as requirements, actions, etc.).

1. Exit out of “Entity View” to go to the “Test Suite.”
2. Click on  , then select “Traceability Matrix.”
3. On the left menu bar, click on the dropdown box under “Top (X-Axis)” and select “Related.”
4. Under the “Relationship Type,” type or scroll down to select “verifies.”
 - o *Note: A matrix should appear that shows if a test case entity has the “verifies” relationship with any other entity in the project. The boxes can be clicked on, and a relationship will be created between those two entities automatically.*
5. Click on  to save the matrix and give the following attributes:
 - a) Number: “MA.1”
 - b) Name: “Test Case Matrix”
 - c) Description: “Traceability Matrix showcasing the verifies relationship.”

Notes on Test Cycles

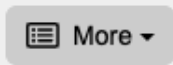
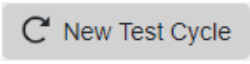
By identifying an issue, program management will know a problem was identified during the verification process. Once a problem has been resolved, you can rerun that portion of the Test Suite via “New Test Cycle” and capture those results as well.

The test case status can be updated under the “Status” bar for each requirement. The “Status Roll-Up” bar will be updated with a visual and counting total of all the statuses that have been received during iterations of testing. A running total will be counted in all parent Test Case levels. An example of what mid-testing looks like for the Rover is in the diagram below.

Entity	Expected Result	Actual Result	Status	Status Roll-Up
TS.2.1 Survey Rover Operational Test and E... Overall final testing for rover in a simulated operational environment.	Meets all operational requirements for Phase 1 lunar survey mission	TBD	Not Run	6
TS.2.2 Survey Rover System Level Verification	Meets all system level requirements for Phase 1 lunar survey rover	TBD	Not Run	5
TS.2.3 Survey Rover Subsystem Level Verifi...	Meets all subsystem level requirements for Phase 1 lunar survey rover	TBD	Not Run	5
TS.2.4 Survey Rover Unit Level Verification	Meets all unit level requirements for Phase 1 lunar survey rover	TBD	Failed	4, 2, 3, 4, 24
TS.2.4.1 Rover Scoop Check the rover scoop to ensure it meets the design requirements.	Meets all scoop design requirements	TBD	Failed	2
TS.2.4.1.1 Rover Scoop Inspection Verify the rover scoop workmanship meets standards.	Scoop constructed with proper materials and finishing.	Rough edges found on left side of scoop	Failed	Failed
TS.2.4.1.2 Rover Scoop Tensile Strength Test Verify the rover scoop meets or exceeds amount of material required for analysis.	Scoop can hold at least 25 kilograms of simulated lunar regolith.	Scoop broke only after reaching 50 kilograms of simulated lunar regolith	Failed	Failed
TS.2.4.2 Rover Cameras Verification	Meets all camera design requirements	TBD	Blocked	1, 1, 1, 4
TS.2.4.3 Rover Sensor Unit Verification	Meets all sensor unit design requirements	All requirements met. See results in attached file	Passed	4
TS.2.4.4 Rover Antennas Verification	Meets all antenna design requirements	All requirements met. See results in attached file	Passed	2
TS.2.4.5 Rover Solar Panels Verification	Meets all solar panel design requirements	TBD	Failed	1, 1, 2, 1

Figure 34: Example of Test Suite for the Survey Rover

Test cases can also be added as scenarios are developed. This is a fluid process and can be edited as necessary.

To run a new iteration of testing with status updates, click on the  dropdown button, then push the  button on the top menu bar. Record the results of testing, adjust your system or system-of-systems until all requirements have been completed, and honor all standards and specifications.

Lab 10. Identify O&S and Disposal Processes

Chapter 10 of the textbook states how systems engineers can play an important role in the Operations and Support process. Systems engineers can observe current operations within a system and identify potential areas for improvement. They can also produce documents and architecture that prove useful to operations and support personnel.

Document an Identified Issue

In this scenario, Operations and Support personnel have identified an issue that they have submitted into the SE database.

1. Navigate to the “Database” from the top Menu tab.
2. Press the  button and create an “Issue” entity.
3. Fill in the following attributes:
 - a) Number: “1.748”
 - b) Name: “Control delay causing rover problems and risks.”
 - c) Description: “The delay in the transmission of signals from the Moon to the Earth has caused some problems with the rover control. The current process needs correction before damage can occur to the rover.”
 - d) Status: “Open”
 - e) Assignee: *Insert your username here*
 - o *Note: Continuing the scenario, the problem that was initially detected by the Operations and Support staff is now your responsibility as the systems engineer. Your task is to monitor the issue, pinpoint possible causes, and suggest solutions.*
 - f) Priority: “Critical”

Add Relationships to Finish Tracking the Issue

After a detailed investigation, you have identified the root cause of the issue and devised mitigation strategies (which carry their own risks). The next step is to complete the tracking of the issue and illustrate the connections by creating relationships.

1. In the same entity, locate the “relationships” section to the right and scroll to “caused by Many”.
2. Press the  button, adding “3.27 Communications Plan” and “AST.ROV.1.7.1 Rover Software.”
3. Scroll to the relationship, “mitigated by Many,” then press the arrow next to the  button to create a new Action.
4. Give the new “Action” entity the following attributes:
 - a) Number: “WA.435”
 - b) Name: “Upgrade software to enhance autonomous operations.”

After completing the previous steps, your screen should look like Figure 35.

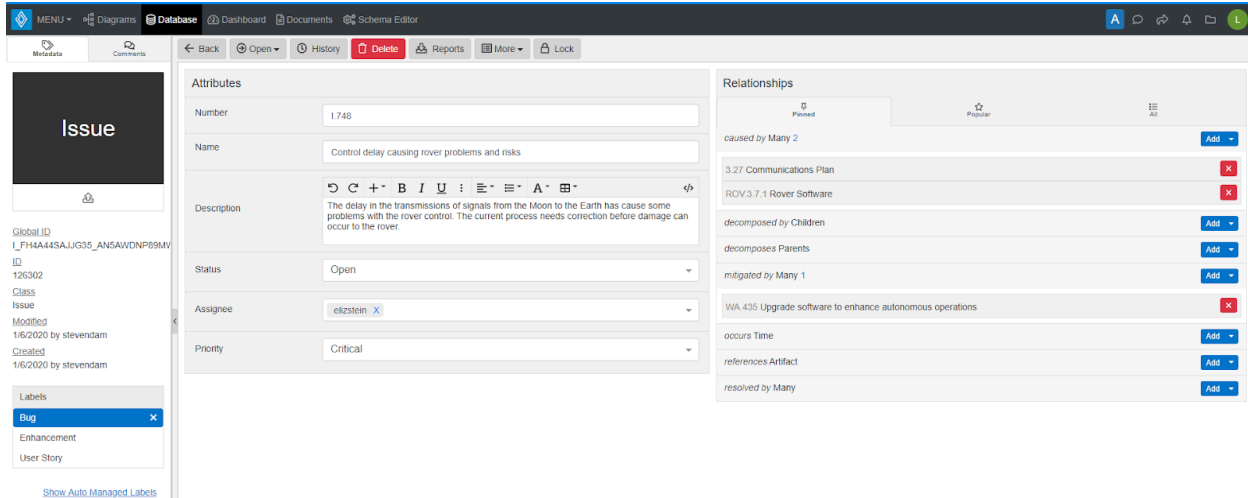
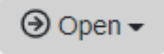


Figure 35. Capturing Issues

5. Press the green “Save” button to save and exit.

Trace Issues via a Spider Diagram

Looking at issue I.748 in a spider diagram allows for visual tracing of issues to causes and mitigations.

1. Click “Database” on the Menu tab.
2. Find the “Issue” class entity, “I.748 Control delay causing rover problems and risks.”
 - *Note: This entity can be found by scrolling through the database, filtering the classes on the left filter menu, or by searching the entity by the search bar on the top.*
3. Select the entity, then click on the  button to open a dropdown list of items.
4. From the dropdown, click on “Spider Diagram” underneath “LML Diagrams.”
 - *Note: Your screen should then look like Figure 36.*

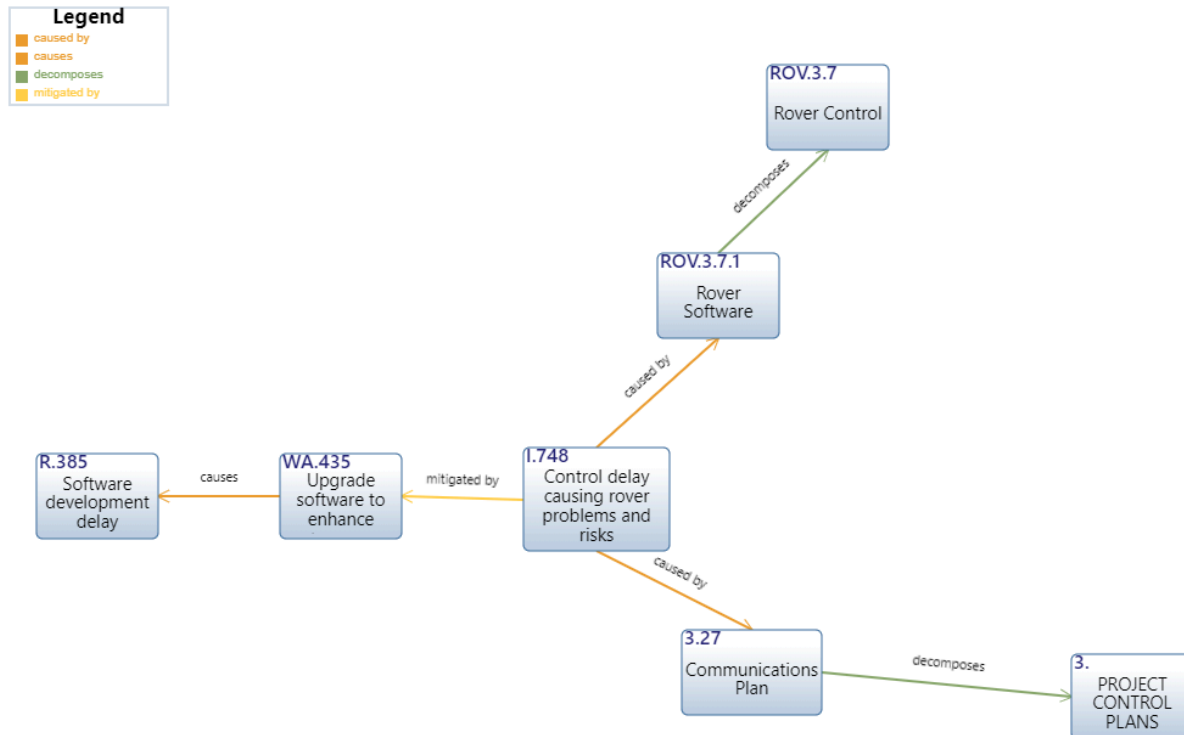
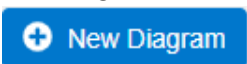


Figure 36. Traceability Via the Spider Diagram

This view makes it easy to see the relationships between these entities and highlights process improvement opportunities from a systems engineer's perspective. For instance, when upgrading rover software to boost performance, it's practical to evaluate concurrent software upgrades to reduce delays. Additionally, this may require greater processing power and increased data storage capacity.

Note on Disposal

As mentioned in the textbook, planning for disposal should be a part of the early systems engineering activities for any system. If a plan has not been developed and stored in the SE database, it will become urgent when a decision is made to dispose of a system (which will occur eventually). The process for disposal begins with a disposal plan which can be captured as an Artifact in Innoslate. However, for the purposes of this lab, the disposal process will be reiterated in the form of an Action diagram in Innoslate.

1. Navigate to "Diagrams View" from the Menu tab.
2. Click on  to create a new Action Diagram.
3. Fill in the following attributes:
 - a) Number: "D."
 - b) Name: "Disposal Action Process"

c) *Description: “This diagram details a generic disposal process that can be tailored for any system.”*

4. *Create the diagram shown below:*

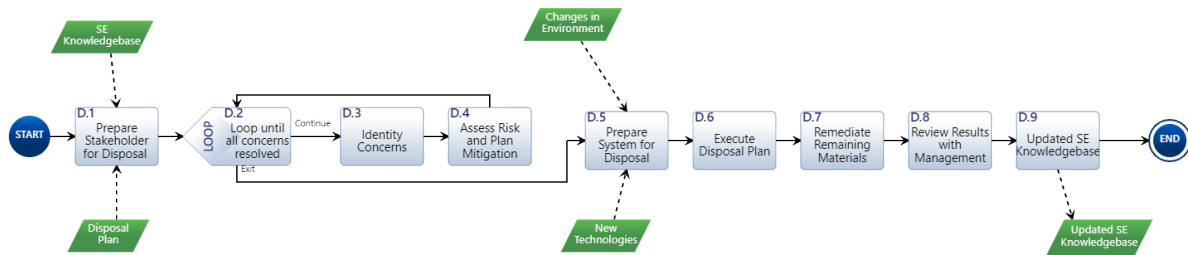


Figure 37. Disposal Action Diagram

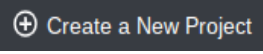
5. *Press the green “Save” button to save and exit.*

Now that you have the disposal process in mind, consider adding it to your project by creating new plans, documents, or other requirements to ensure that the end of life for your project is planned out.

Lab 11. Conduct MBR of Other Teams Work

Begin your MBR (Model Based Review) for this semester's labs. First, you will need to create a new project which the classmate reviewing your project can use to grade.

Create and Share a New Project

1. While in a project, click the "Switch Projects" () button in the top right corner of Innoslate's toolbar to open a drop-down menu.
2. Click on "Create a New Project" ().
3. Fill in the following attributes:
 - a) Name: "*Your Name* + Model Based Review"
 - b) Description: "*Your Name's* Model Based Review project which *Classmate's Name* will be reviewing."
 - o *Note: Replace the italicized parts with your name and the name of the classmate who will be grading your project. Your professor will inform you of which folder the project should be placed in.*
4. Click "Create & Go."

Link Evaluation Criteria to the Project

Update each of the evaluation criteria by clicking on the statement and adding a hyperlink to the described activity in your original project.

1. Go to the "Test Center" from the top menu bar.
 - o *Note: This test suite will be provided to you. If it is not there, contact your professor.*
2. Click into the "Evaluation Criteria" Test Suite.
 - o *Note: Here you will see the content you are being graded on for each criteria under the Expected Result column.*
3. Select the "Evaluation Criteria Statement."
4. Click the "+" button then "Link."
5. Add the URL and Name of the document, diagram, entity, class, label, or whatever else you may be adding for review.
 - o *For example, 1.1 Project Management Plan would be complete like so:*

Insert Link
✕

URL

https://cloud.innoslate.com/spec/p/470/documents/126397

Text

PMP.1 Project Management Plan

Cancel
Insert

EC Evaluation Criteria	Expected Result
- 1 LAB 1	
- 1.1 PROJECT MANAGEMENT PLAN Put a hyperlink to your Project Management Plan Document here. PMP.1 Project Management Plan	The Project Management Plan for Moonbase: Selene has been uploaded.

Figure 38. PMP Attributes

6. Repeat this process for every statement in the “Evaluation Criteria” document.

Baseline Documents for Review

Baselining documents before allowing others to review is necessary to track any changes made.

1. From the “Documents View,” select “R.1 Top-Level Requirements Document for Selene.”
2. Click the dropdown button and select “Baseline.”
3. Name this baseline, “Pre-MBR Requirements Document.”
4. Select the blue “Create” button. The document is now baselined, turning the yellow bar to blue.
 - o *Note: This action cannot be undone, as organizations will use this as part of a legal process to ensure the integrity of documentation. Diagrams, however, cannot be baselined. To include a model in the review, hyperlink it to the appropriate section in the document being reviewed using the “Diagram” option, as shown below.*

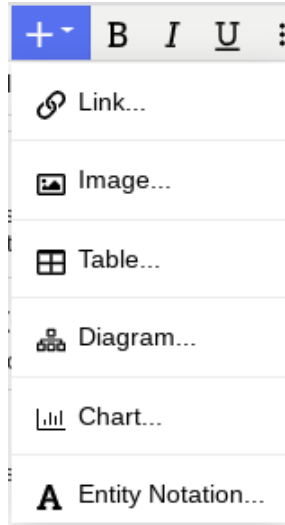


Figure 39. “Add” Button Functions

Add Reviewers to the Project

Once the documents and models have been hyperlinked and baselined, you can now provide the reviewers access to your “Moonbase: Selene” project. With “read-only” access, your classmates may be able to just view your hyperlinks and navigate your project; they may post comments in the document and diagram views or directly in the individual entity views.

1. Click the Share () button in the top-right corner of Innoslate’s display to enter “Project Permissions.”
2. Type in your peer’s Innoslate username or email associated with their Innoslate account, and select “Reviewer” from the Role drop-down menu.
3. Click the blue “Share” button once you are finished.
 - o Note: A “viewer” can **only** see your project while a “reviewer” can see your project **and** make comments. Those added to your project must have an Innoslate account to view any project. You can change the share status of each person in your project at any point, as well as remove their access, by returning to the “Project Permissions” window.

Add Collaborator Access for Evaluating

Next, share your Model-Based Review project with the classmate who will be evaluating your project. With this permission, they will be able to grade your MBR project by changing the status of each Evaluation Criteria.

1. Follow steps 1-3 from above, but now select “Collaborator” for the role.
 - o Note: The selected individuals will receive an email from Innoslate when they have been granted permission to access another Innoslate project. They can also access this project at any time on the Dashboard.

Add a Comment to a Document

1. While in the document you wish to review, click on the desired entity.
2. On the left-side panel, go to the “Comments” tab.
3. Type your comment and click  to submit it.
 - o *Note: The project owner will receive an email notifying them of your comment. You can also resolve and reopen your comments, as well as reply to others’ comments by selecting it.*

Closing Notes

As the project owner, you may view the comments made on your project at any time by going to the

Dashboard and clicking the  button to add the “Comments Widget.”

Once you are ready to begin grading your classmate’s “Moonbase: Selene” project, go to the Model-Based Review project you have been granted access to as a Collaborator. Go to the “Evaluation Criteria Test Suite” and begin reviewing the project by clicking each hyperlinked Entity.

If your classmate has presented and completed the appropriate Entity, click on the “Status” column of that Evaluation Criteria and change the status to “Passed.” If your classmate is lacking the appropriate Entity, mark the status as “Failed” and state what is wrong with this hyperlinked Entity in the “Actual Results” Column. If you marked any as failed, let your classmate know so they may review your comments and change their project as necessary. Once they are done, grade their changes and update the status of the criteria.

The grader will submit the Test Suite results to your professor. To do this, click  at the top on the “Evaluation Criteria Test Suite.” Select “Test Cases Output (DOCX)” and name it appropriately.

Congratulations! You have successfully completed the Lab Manual for Real MBSE!